CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Chandubhai S. Patel Institute of Technology (CSPIT)

K.D. Patel Department of Information Technology • Academic Year 2026–2027

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DEPT : CSPIT	**SUBJECT :** OOP (CEUE203)	**Semester :** 3rd Semester

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Total Practicals Completed: 14 / 14 **Faculty In-Charge Signature:** _____ **Date:** _____

PRACTICAL 1-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 1 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Vending Machine, Toll Booth, RPSLS and Part B: MiniBank menu shell.

CODE SCREENSHOT:

RPSLS.java 1

Part A > Practical 1-A > RPSLS.java > ...

```java
import java.util.*;

public class RPSLS {
    enum Move {
        ROCK, PAPER, SCISSORS, LIZARD, SPOCK
    }

    static int winner(Move a, Move b) {
        if (a == b)
            return 0;
        boolean beats = switch (a) {
            case ROCK -> b == Move.SCISSORS || b == Move.LIZARD;
            case PAPER -> b == Move.ROCK || b == Move.SPOCK;
            case SCISSORS -> b == Move.PAPER || b == Move.LIZARD;
            case LIZARD -> b == Move.SPOCK || b == Move.PAPER;
            case SPOCK -> b == Move.SCISSORS || b == Move.ROCK;
        };
        return beats ? 1 : -1;
    }

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Random random = new Random();
        int player = 0, computer = 0;
        for (int i = 1; i <= 5; i++) {
            System.out.print("Round " + i + " (ROCK/PAPER/SCISSORS/LIZARD/SPOCK): ");
            Move p = Move.valueOf(sc.next().toUpperCase());
            Move c = Move.values()[random.nextInt(Move.values().length)];
            int result = winner(p, c);
            if (result > 0)
                player++;
            else if (result < 0)
                computer++;
            System.out.println("You: " + p + ", Computer: " + c + " -> "
                    + (result == 0 ? "Tie" : result > 0 ? "You win" : "Computer wins"));
        }
        System.out.println(player > computer ? "You win " + player + "" + computer
                : player < computer ? "Computer wins " + computer + "" + player
                        : "Overall tie " + player + "" + computer);
    }
}

```

Code Screenshot 1

TollBooth.java 1

Part A > Practical 1-A > TollBooth.java > ...

```java
import java.util.*;

public class TollBooth {
    record Vehicle(String number, String type) {
    }

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int total = 0, bike = 0, car = 0, truck = 0;
        while (true) {
            System.out.print("Vehicle number (done to finish): ");
            String number = sc.next().trim();
            if (number.equalsIgnoreCase("done"))
                break;
            System.out.print("Type (bike/car/truck): ");
            String type = sc.next().trim().toLowerCase();
            Vehicle v = new Vehicle(number, type);
            int toll = switch (v.type()) {
                case "bike" -> 20;
                case "car" -> 50;
                case "truck" -> 150;
                default -> 0;
            };
            if (toll == 0) {
                System.out.println("Unknown vehicle type");
                continue;
            }
            total += toll;
            switch (type) {
                case "bike" -> bike++;
                case "car" -> car++;
                case "truck" -> truck++;
            }
        }
        String frequent = bike >= car && bike >= truck ? "bike" : car >= truck ? "car" : "truck";
        System.out.println("Total toll: " + total);
        System.out.println("Most frequent: " + frequent);
    }
}

```

Code Screenshot 2

VendingMachine.java 1

Part A > Practical 1-A > VendingMachine.java > VendingMachine > main(String[])

```java
import java.util.Scanner;

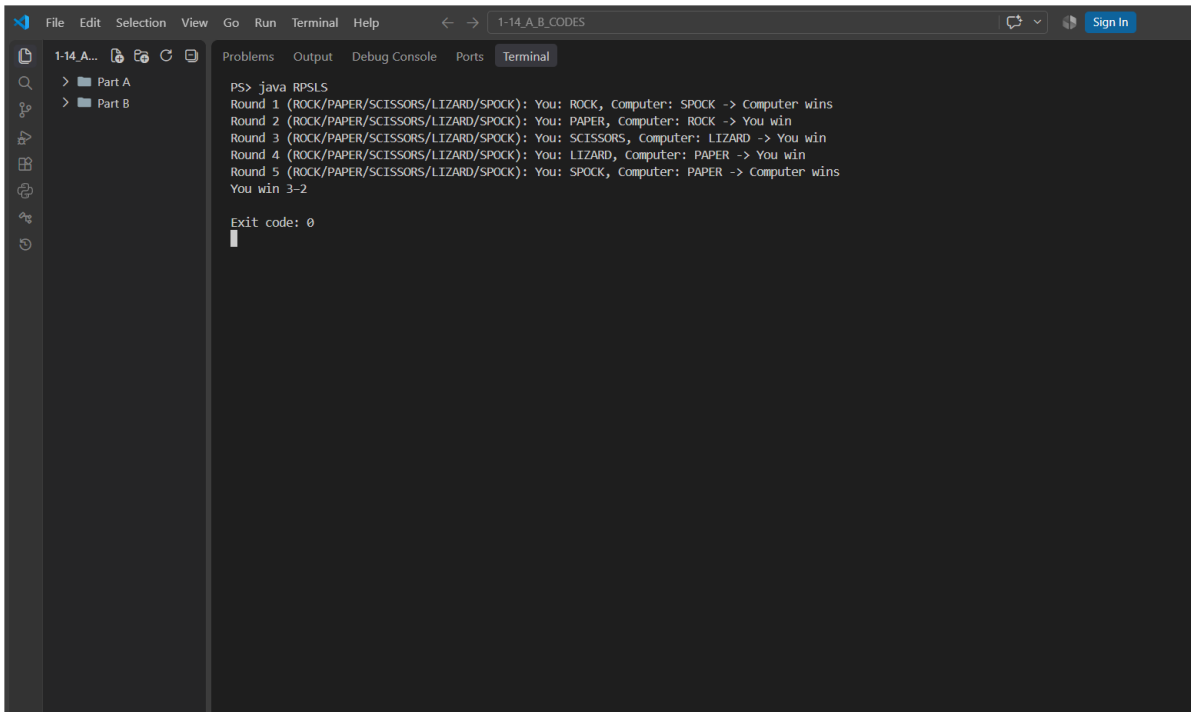
public class VendingMachine {
    enum Coin {
        ONE, TWO, FIVE, TEN
    }

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int total = 0, price = 15;
        while (total < price) {
            System.out.print("Enter coin (ONE, TWO, FIVE, TEN): ");
            Coin coin = Coin.valueOf(sc.next().trim().toUpperCase());
            total += switch (coin) {
                case ONE -> 1;
                case TWO -> 2;
                case FIVE -> 5;
                case TEN -> 10;
            };
            System.out.println("Total so far: " + total);
        }
        System.out.println("Paid. Change: " + (total - price));
    }
}

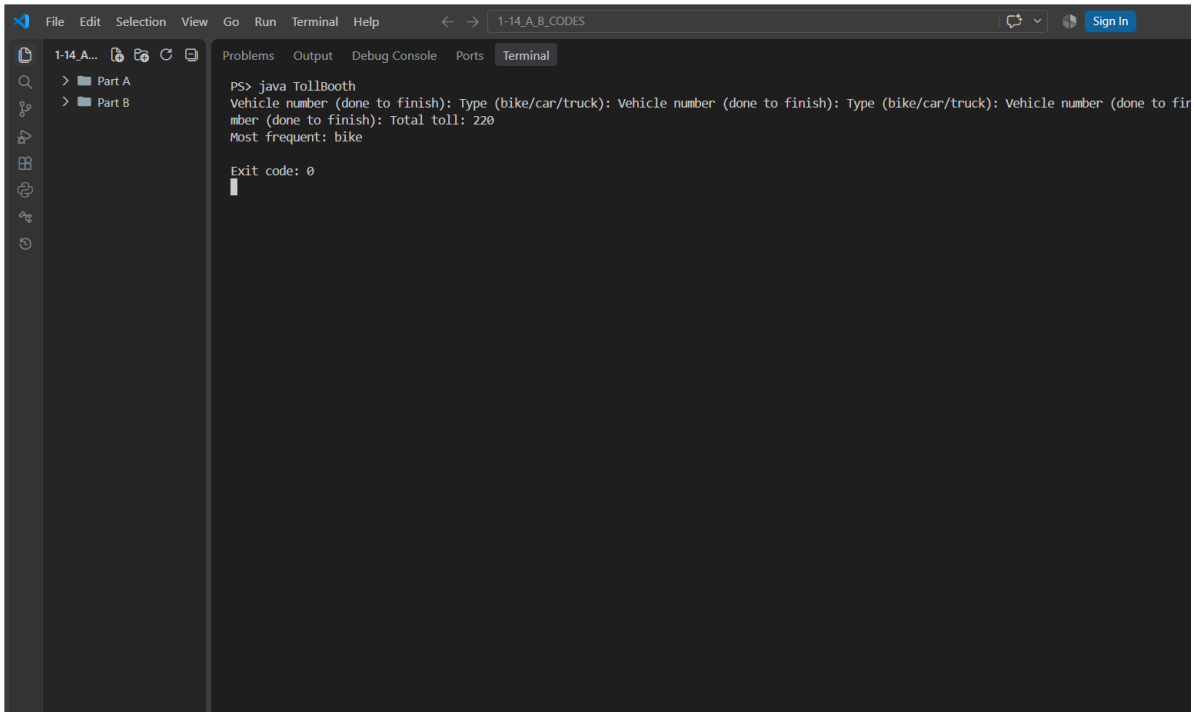
```

Code Screenshot 3

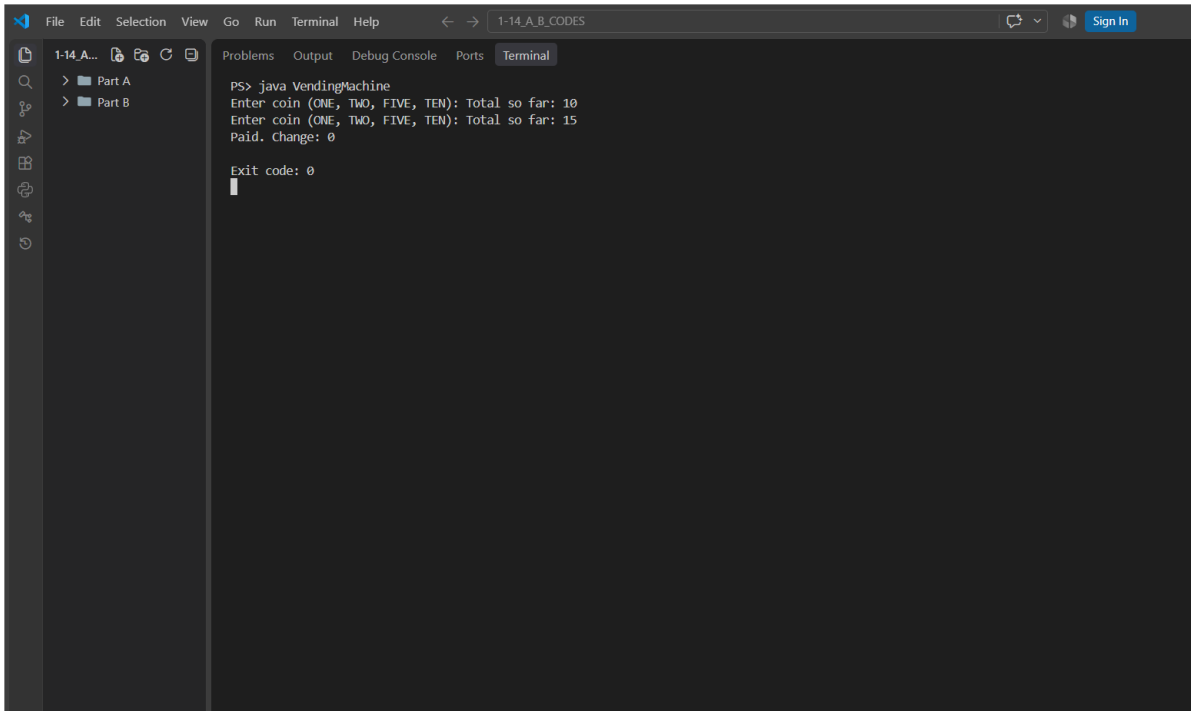
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Develop a basic interactive Java console application and use enums, records, switch expressions, and input handling.

PRACTICAL 2-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 2 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Thermostat, Cinema Show, Parking Lot and Part B: Customer and Account encapsulation.

CODE SCREENSHOT:

CinemaShow.java

Part A > Practical 2-A > CinemaShow.java > ...

```java
public class CinemaShow {
    private String title;
    private int seatsAvailable;
    private final int capacity;
    private static int totalBooked = 0;

    public CinemaShow(String title, int capacity) {
        this.title = title;
        this.capacity = capacity;
        seatsAvailable = capacity;
    }

    public CinemaShow(String title) {
        this(title, capacity: 100);
    }

    public boolean book(int n) {
        if (n > 0 && n <= seatsAvailable) {
            seatsAvailable -= n;
            totalBooked += n;
            return true;
        }
        return false;
    }

    public void cancel(int n) {
        if (n > 0)
            seatsAvailable = Math.min(capacity, seatsAvailable + n);
    }

    public int getSeatsAvailable() {
        return seatsAvailable;
    }

    public static int getTotalBooked() {
        return totalBooked;
    }

    Run | Debug
    public static void main(String[] args) {
        CinemaShow show = new CinemaShow(title: "Java Premiere", capacity: 50);
        System.out.println("Book 20: " + show.book(n: 20) + ", seats: " + show.getSeatsAvailable());
        System.out.println("Book 40: " + show.book(n: 40) + ", seats: " + show.getSeatsAvailable());
        show.cancel(n: 5);
        System.out.println("After cancellation, seats: " + show.getSeatsAvailable());
        System.out.println("Total booked: " + getTotalBooked());
    }
}
```

Code Screenshot 1

ParkingLot.java

Part A > Practical 2-A > ParkingLot.java > ...

```java
public class ParkingLot {
    private int twoWheelers = 0, fourWheelers = 0;
    private final int twoCap, fourCap;
    private static long revenue = 0;

    public ParkingLot(int twoCap, int fourCap) {
        this.twoCap = twoCap;
        this.fourCap = fourCap;
    }

    public boolean park(String type) {
        if (type.equalsIgnoreCase("two") && twoWheelers < twoCap) {
            twoWheelers++;
            revenue += 20;
            return true;
        }
        if (type.equalsIgnoreCase("four") && fourWheelers < fourCap) {
            fourWheelers++;
            revenue += 40;
            return true;
        }
        System.out.println("Full");
        return false;
    }

    public void leave(String type) {
        if (type.equalsIgnoreCase("two") && twoWheelers > 0)
            twoWheelers--;
        else if (type.equalsIgnoreCase("four") && fourWheelers > 0)
            fourWheelers--;
    }

    Run | Debug
    public static void main(String[] args) {
        ParkingLot p = new ParkingLot(twoCap: 2, fourCap: 1);
        p.park(type: "two");
        p.park(type: "four");
        p.park(type: "two");
        p.park(type: "two");
        p.leave(type: "two");
        System.out.println(
                "Two-wheelers: " + p.twoWheelers + ", Four-wheelers: " + p.fourWheelers + ", Revenue: " + revenue);
    }
}

```

Code Screenshot 2

Thermostat.java 1

Part A > Practical 2-A > Thermostat.java > ...

```java
public class Thermostat {
    private String location;
    private int temperature;
    private static final int MIN = 16, MAX = 30;
    private static int activeCount = 0;

    public Thermostat(String location, int startTemp) {
        this.location = location;
        temperature = (startTemp >= MIN && startTemp <= MAX) ? startTemp : 22;
        activeCount++;
    }

    public Thermostat(String location) {
        this(location, startTemp: 22);
    }

    public void raise() {
        if (temperature < MAX)
            temperature++;
        else
            System.out.println("Already at maximum (30)");
    }

    public void lower() {
        if (temperature > MIN)
            temperature--;
        else
            System.out.println("Already at minimum (16)");
    }

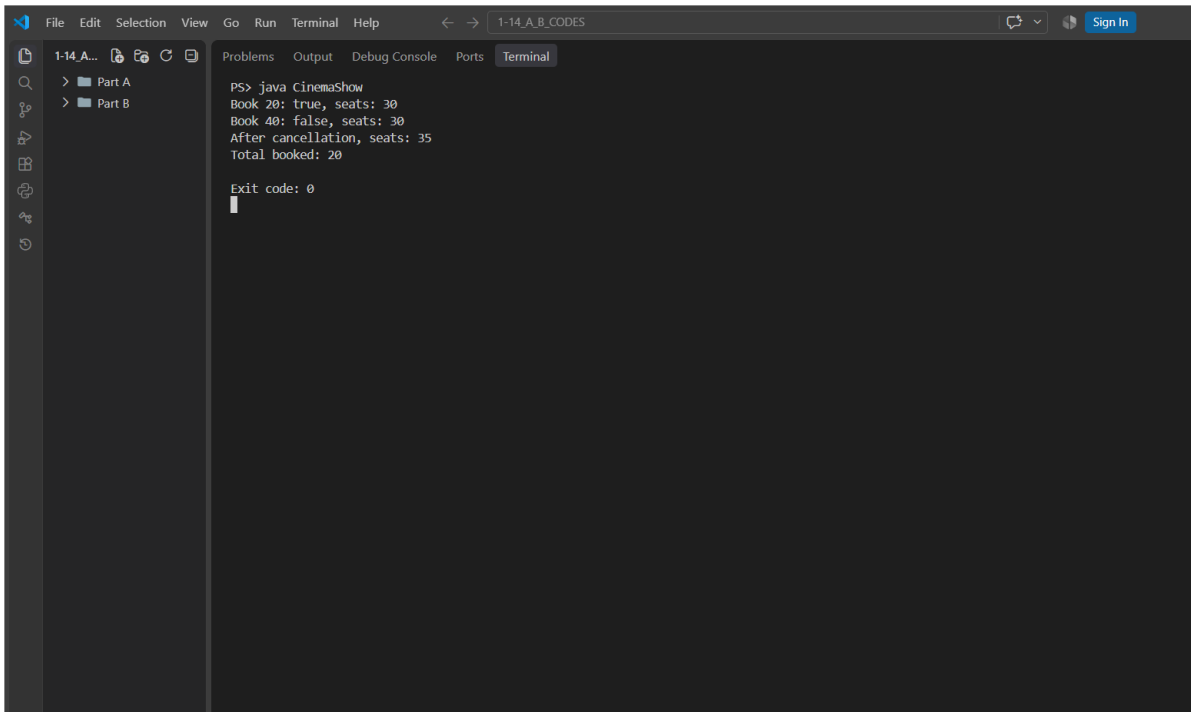
    public int getTemperature() {
        return temperature;
    }

    public static int getActiveCount() {
        return activeCount;
    }

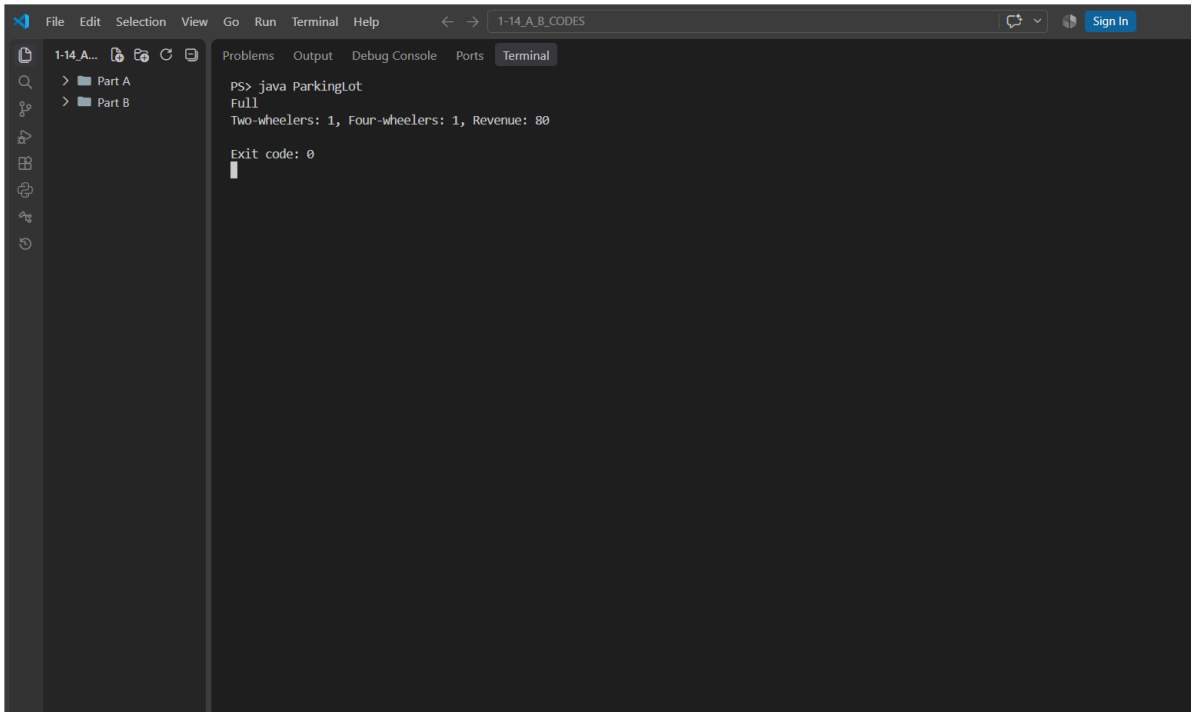
    Run | Debug
    public static void main(String[] args) {
        Thermostat a = new Thermostat(location: "Lab", startTemp: 20), b = new Thermostat(location: "Office");
        for (int i = 0; i < 10; i++) {
            a.raise();
            System.out.println(a.location + ": " + a.getTemperature());
        }
        for (int i = 0; i < 20; i++) {
            a.lower();
            System.out.println(a.location + ": " + a.getTemperature());
```

Code Screenshot 3

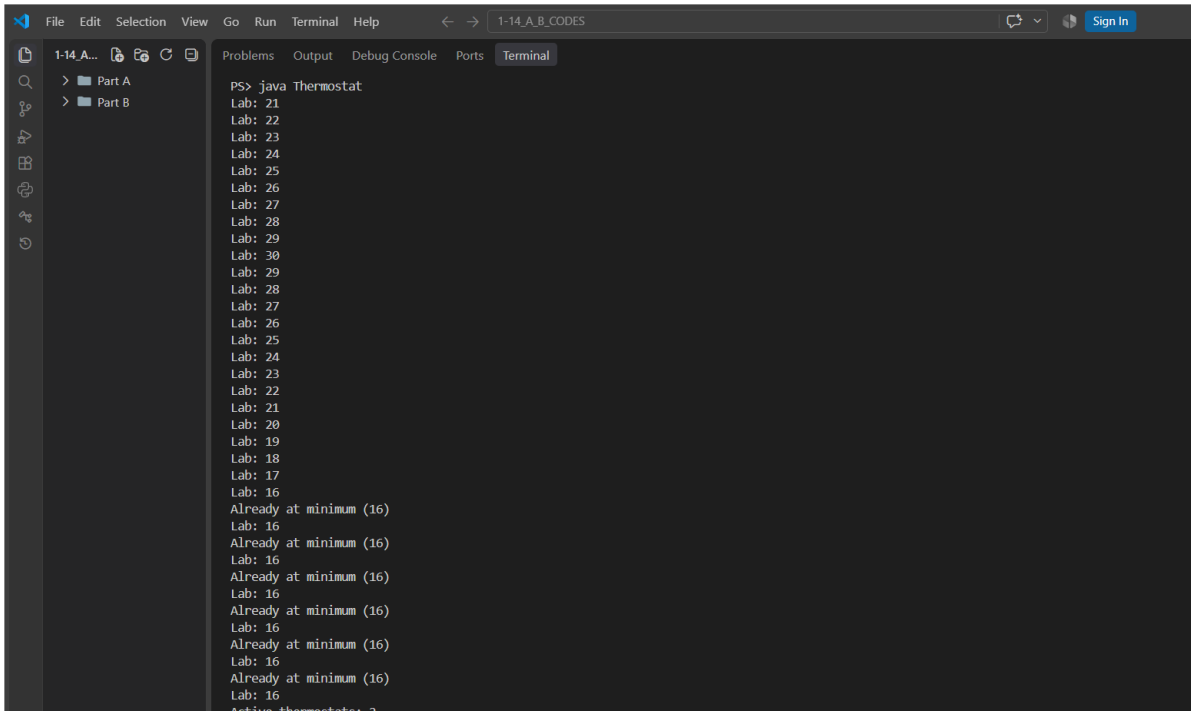
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Apply classes, constructors, encapsulation, static members, and object arrays to model real-world entities.

PRACTICAL 3-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 3 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Distinct Points, Duplicate Cards, Fractions and Part B: Object methods and nested Address.

CODE SCREENSHOT:

Card.java

Part A > Practical 3-A > Card.java > ...

```java
import java.util.Objects;

public class Card {
    private final String rank, suit;
nshul\Downloads\OOP_Practicals_1-13_A_B\Part A\Practical 3-A\Card.java
    t) {
        this.rank = rank;
        this.suit = suit;
    }

    @Override
    public String toString() {
        return rank + " of " + suit;
    }

    @Override
    public boolean equals(Object o) {
        return o instanceof Card c && rank.equals(c.rank) && suit.equals(c.suit);
    }

    @Override
    public int hashCode() {
        return Objects.hash(rank, suit);
    }
}
```

Code Screenshot 1

CardDriver.java

Part A > Practical 3-A > CardDriver.java > ...

```java
import java.util.*;

public class CardDriver {
    Run | Debug
    public static void main(String[] args) {
        Card[] cards = { new Card(rank: "Ace", suit: "Spades"), new Card(rank: "King", suit: "Hearts"), new Card(rank: "Ace", suit: "Spades") };
        List<Card> seen = new ArrayList<>();
        for (Card c : cards) {
            if (seen.contains(c)) {
                System.out.println("Duplicate found: " + c);
                break;
            }
            seen.add(c);
        }
    }
}
```

Code Screenshot 2

Driver.java

Part A > Practical 3-A > Driver.java > ...

```java
import java.util.*;

public class Driver {
    Run | Debug
    public static void main(String[] args) {
        Point[] points = { new Point(x: 1, y: 2), new Point(x: 2, y: 3), new Point(x: 1, y: 2), new Point(x: 4, y: 5), new Point(x: 2, y: 3) };
        List<Point> distinct = new ArrayList<>();
        for (Point p : points)
            if (!distinct.contains(p))
                distinct.add(p);
        System.out.println("Distinct: " + distinct.size());
    }
}

```

Code Screenshot 3

Fraction.java

Part A > Practical 3-A > Fraction.java > ...

```java
import java.util.Objects;

public class Fraction {
    private final int num, den;

    public Fraction(int num, int den) {
        if (den == 0)
            throw new IllegalArgumentException("Denominator cannot be zero");
        if (den < 0) {
            num = -num;
            den = -den;
        }
        int g = gcd(Math.abs(num), den);
        this.num = num / g;
        this.den = den / g;
    }

    private static int gcd(int a, int b) {
        while (b != 0) {
            int t = a % b;
            a = b;
            b = t;
        }
        return a == 0 ? 1 : a;
    }

    @Override
    public String toString() {
        return num + "/" + den;
    }

    @Override
    public boolean equals(Object o) {
        return o instanceof Fraction f && num == f.num && den == f.den;
    }

    @Override
    public int hashCode() {
        return Objects.hash(num, den);
    }

    Run | Debug
    public static void main(String[] args) {
        Fraction a = new Fraction(num: 2, den: 4), b = new Fraction(num: 1, den: 2);
        System.out.println(a);
        System.out.println("Equal: " + a.equals(b));
    }
}
```

Code Screenshot 4

Point.java

Part A > Practical 3-A > Point.java > ...

```java
import java.util.Objects;

public class Point {
    private final int x, y;

    public Point(int x, int y) {
        this.x = x;
        this.y = y;
    }

    @Override
    public String toString() {
        return "(" + x + ", " + y + ")";
    }

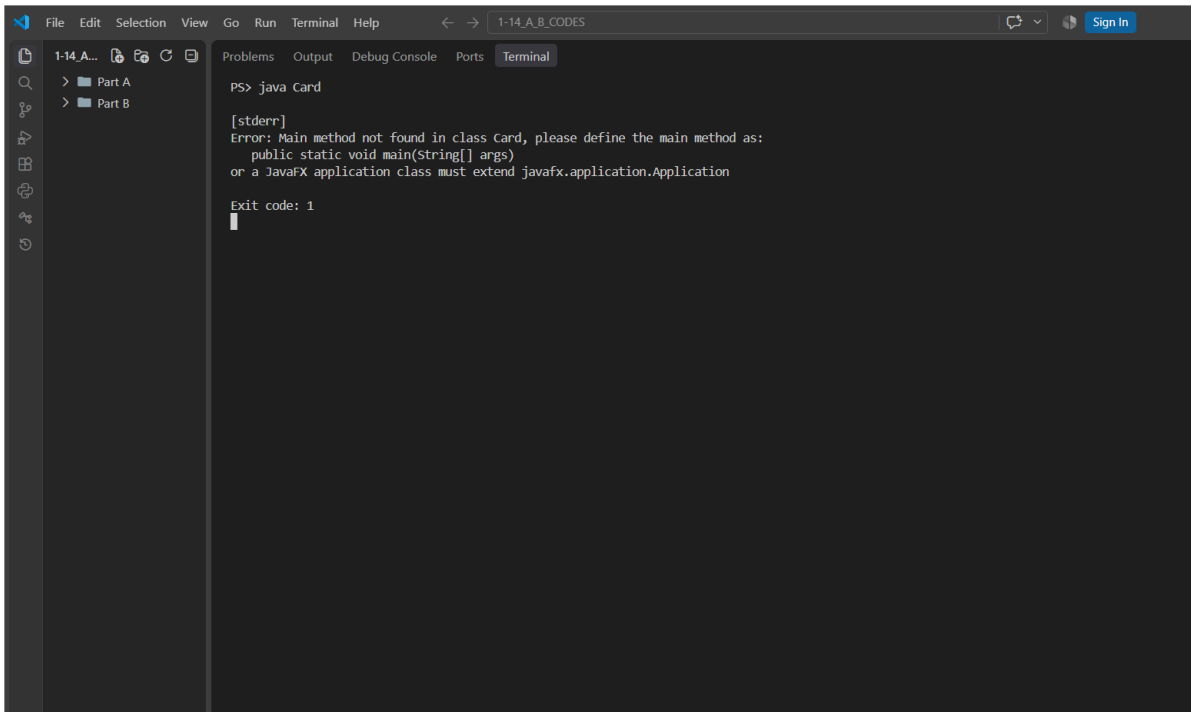
    @Override
    public boolean equals(Object o) {
        return o instanceof Point p && x == p.x && y == p.y;
    }

    @Override
    public int hashCode() {
        return Objects.hash(x, y);
    }
}

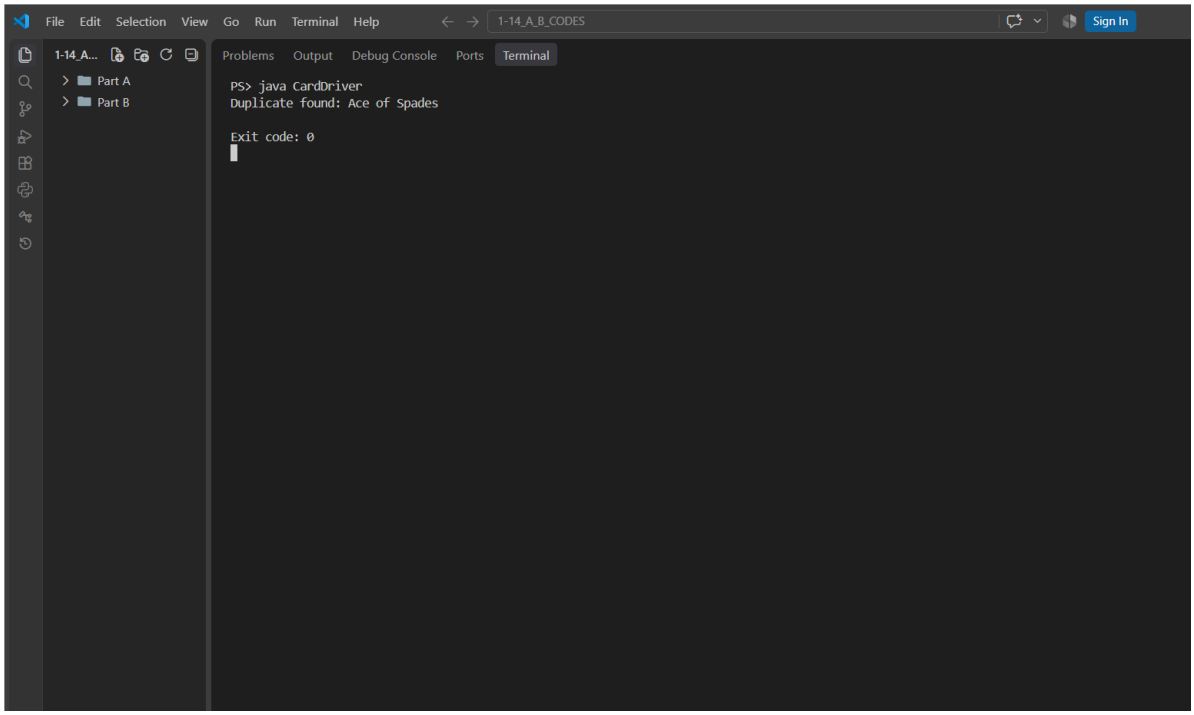
```

Code Screenshot 5

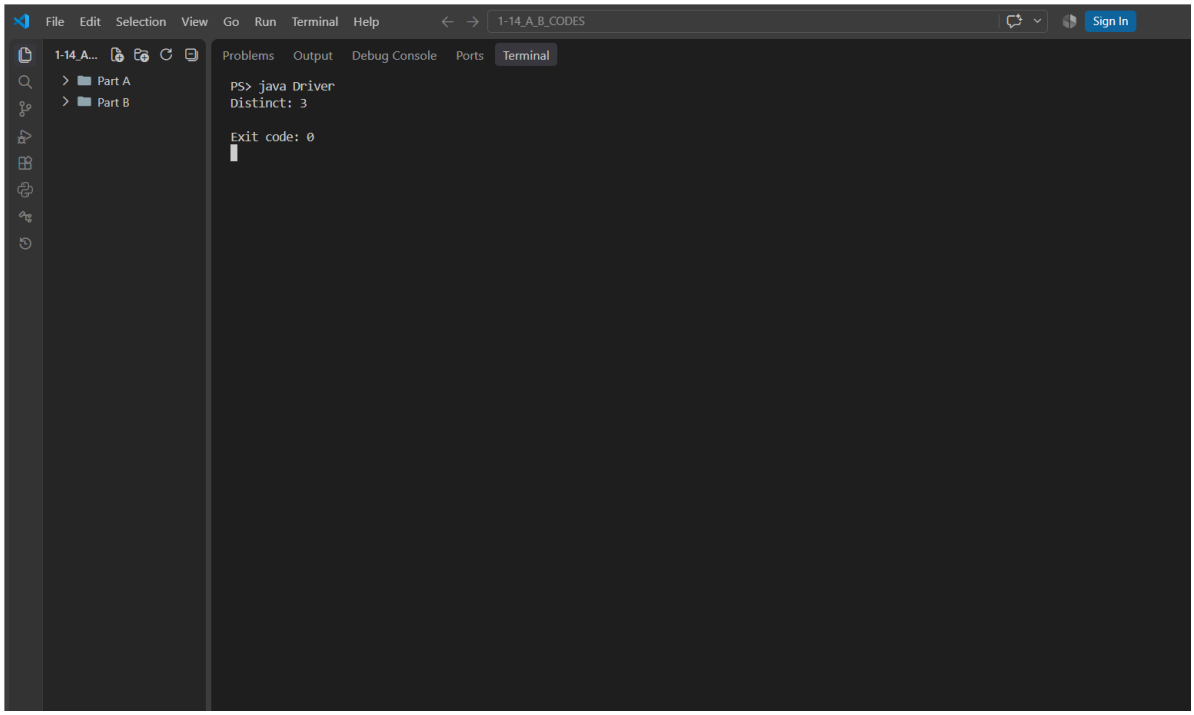
OUTPUT SCREENSHOT:



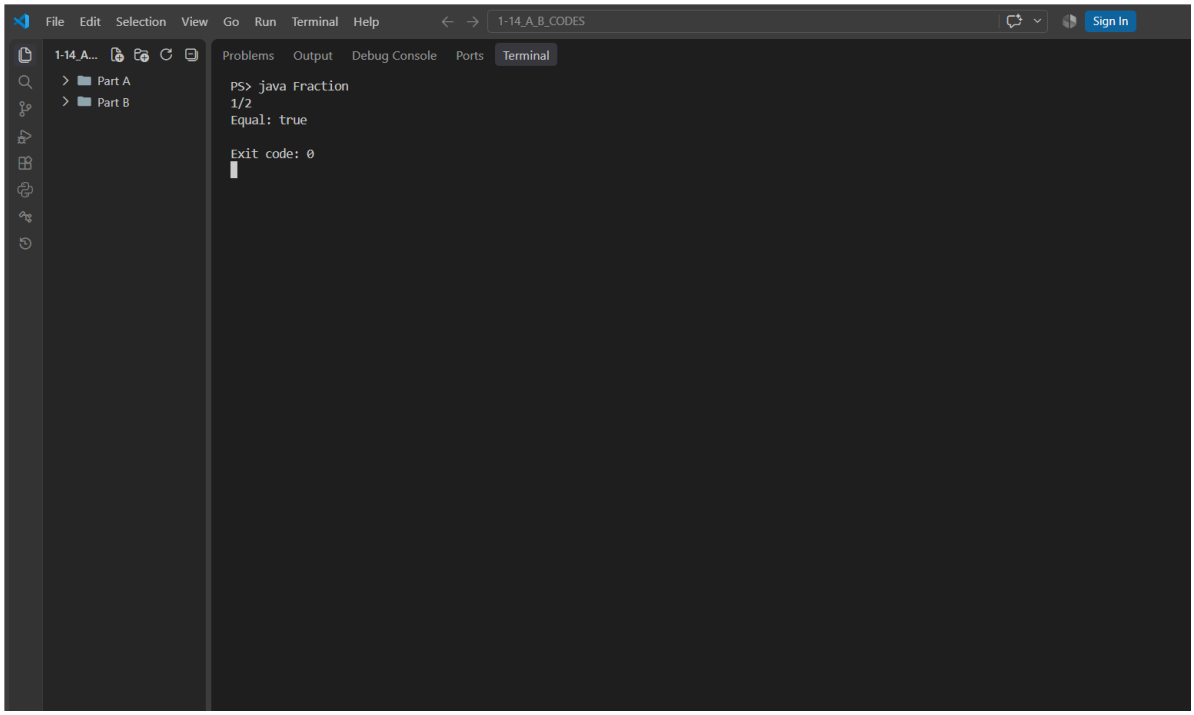
Output Screenshot 1



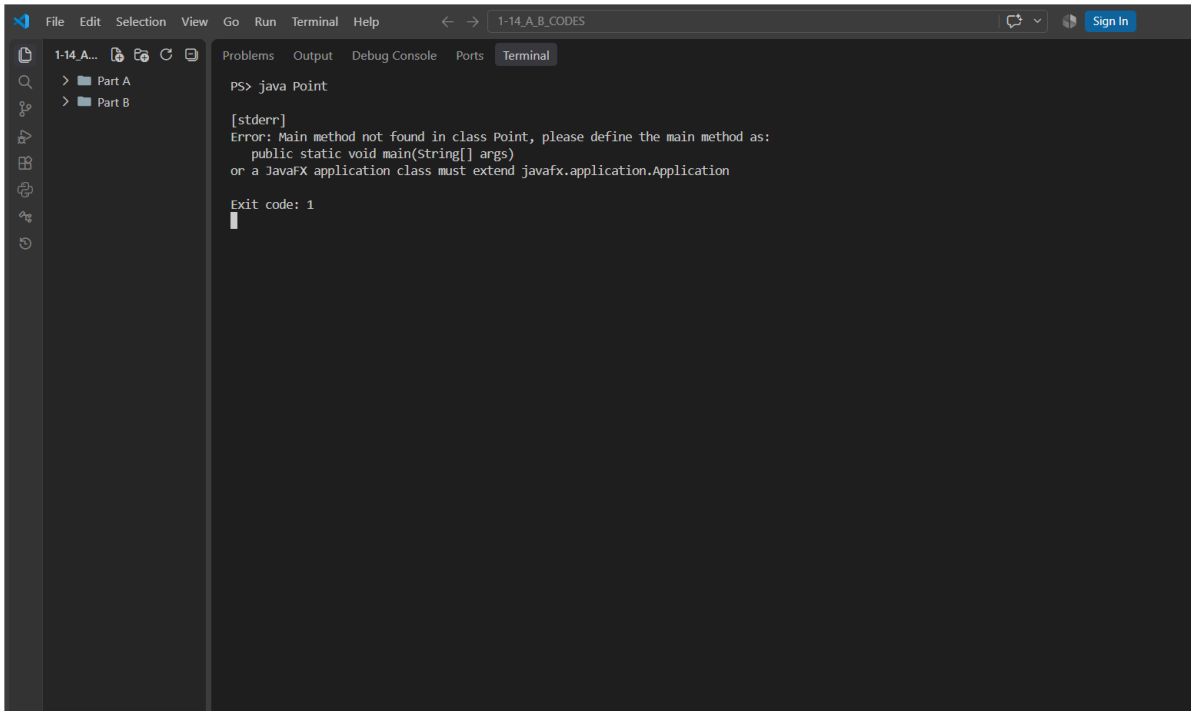
Output Screenshot 2



Output Screenshot 3



Output Screenshot 4



Output Screenshot 5

LEARNING OUTCOMES:

Override Object methods correctly and use equality, hashing, nested classes, and object identity checks.

PRACTICAL 4-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 4 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Password Checker, Chat Filter, Template Filler and Part B: Validators, Command Parser, Statement Formatter.

CODE SCREENSHOT:

ChatFilter.java 1

Part A > Practical 4-A > ChatFilter.java > ...

```java
import java.util.*;

public class ChatFilter {
    Run | Debug
    public static void main(String[] args) {
        String[] logs = { "10:05 alice Hello there", "10:06 bob How are you", "malformed", "10:07 cara HELLO again" };
        Scanner sc = new Scanner(System.in);
        System.out.print("Keyword: ");
        String key = sc.nextLine().toLowerCase();
        StringBuilder report = new StringBuilder();
        int matches = 0;
        for (String line : logs) {
            String[] parts = line.split(" ", 3);
            if (parts.length < 3)
                continue;
            if (parts[2].toLowerCase().contains(key)) {
                matches++;
                report.append(parts[0]).append(" ").append(parts[1]).append(": ").append(parts[2])
                        .append(System.lineSeparator());
            }
        }
        System.out.println("Matches: " + matches);
        System.out.print(report);
    }
}

```

Code Screenshot 1

PasswordChecker.java

Part A > Practical 4-A > PasswordChecker.java > ...

```java
public class PasswordChecker {
    public static String strength(String pw) {
        int score = 0;
        if (pw.length() >= 8)
            score++;
        if (pw.matches(".*[A-Z].*"))
            score++;
        if (pw.matches(".*\\d.*"))
            score++;
        if (pw.matches(".*[^A-Za-z0-9].*"))
            score++;
        return score <= 1 ? "Weak" : score <= 3 ? "Medium" : "Strong";
    }

    Run | Debug
    public static void main(String[] args) {
        for (String p : new String[] { "abc", "Abcd1234!", "hello123", "HELLO!!" })
            System.out.println(p + " -> " + strength(p));
    }
}

```

Code Screenshot 2

TemplateFiller.java

Part A > Practical 4-A > TemplateFiller.java > ...

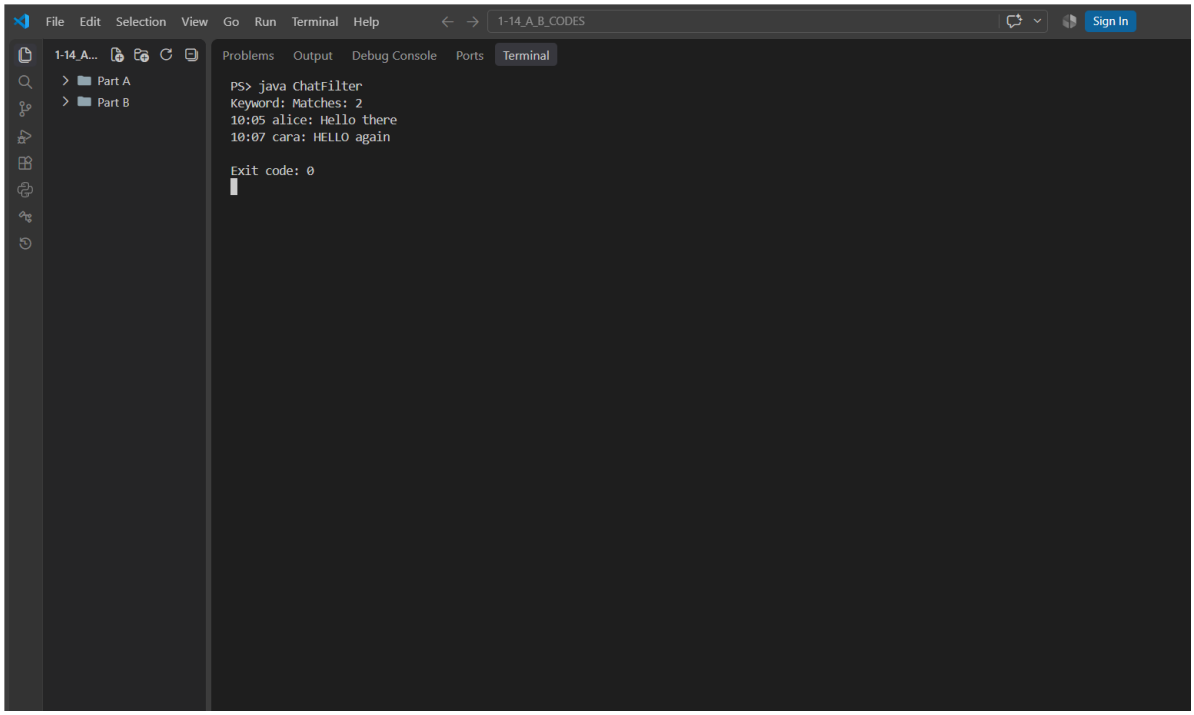
```java
import java.util.regex.*;

public class TemplateFiller {
    Run | Debug
    public static void main(String[] args) {
        String template = "Dear {name}, order {id} ships {date}.";
        String[] names = { "name", "id" }, values = { "Riya", "A07" };
        Matcher m = Pattern.compile("\\{(\\w+)\\}").matcher(template);
        StringBuilder out = new StringBuilder();
        int last = 0;
        while (m.find()) {
            out.append(template, last, m.start());
            String key = m.group(1), value = "[?]";
            for (int i = 0; i < names.length; i++)
                if (names[i].equals(key))
                    value = values[i];
            out.append(value);
            last = m.end();
        }
        out.append(template.substring(last));
        System.out.println(out);
    }
}

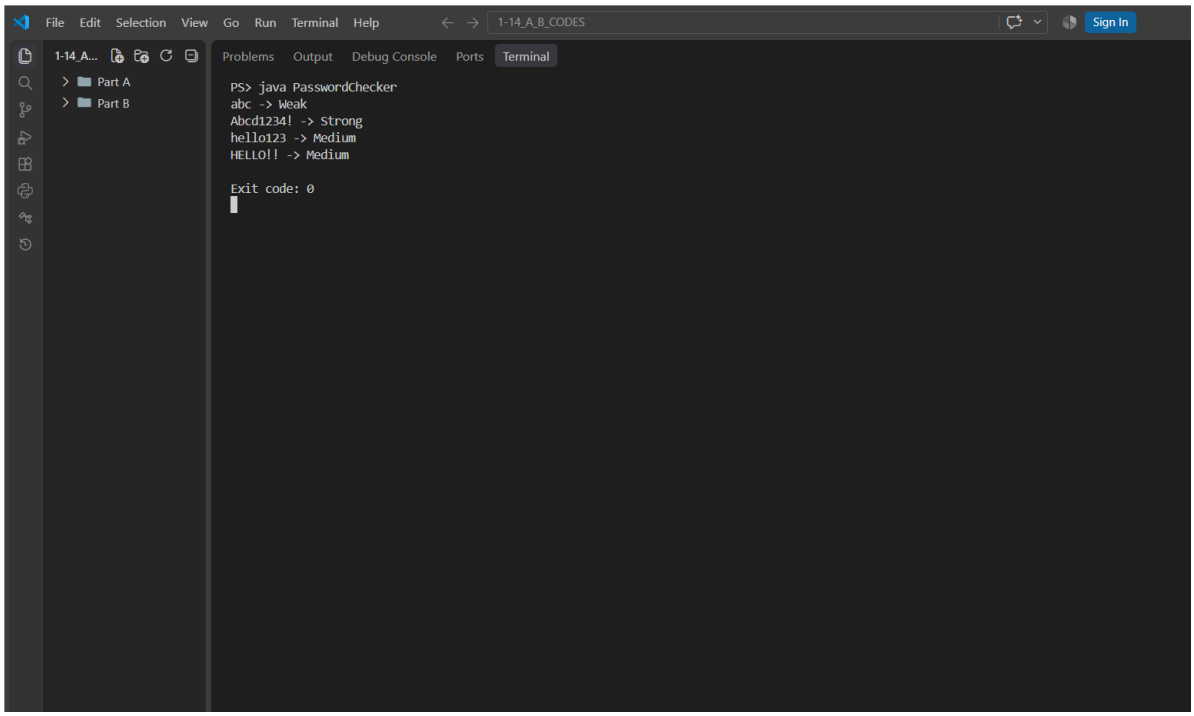
```

Code Screenshot 3

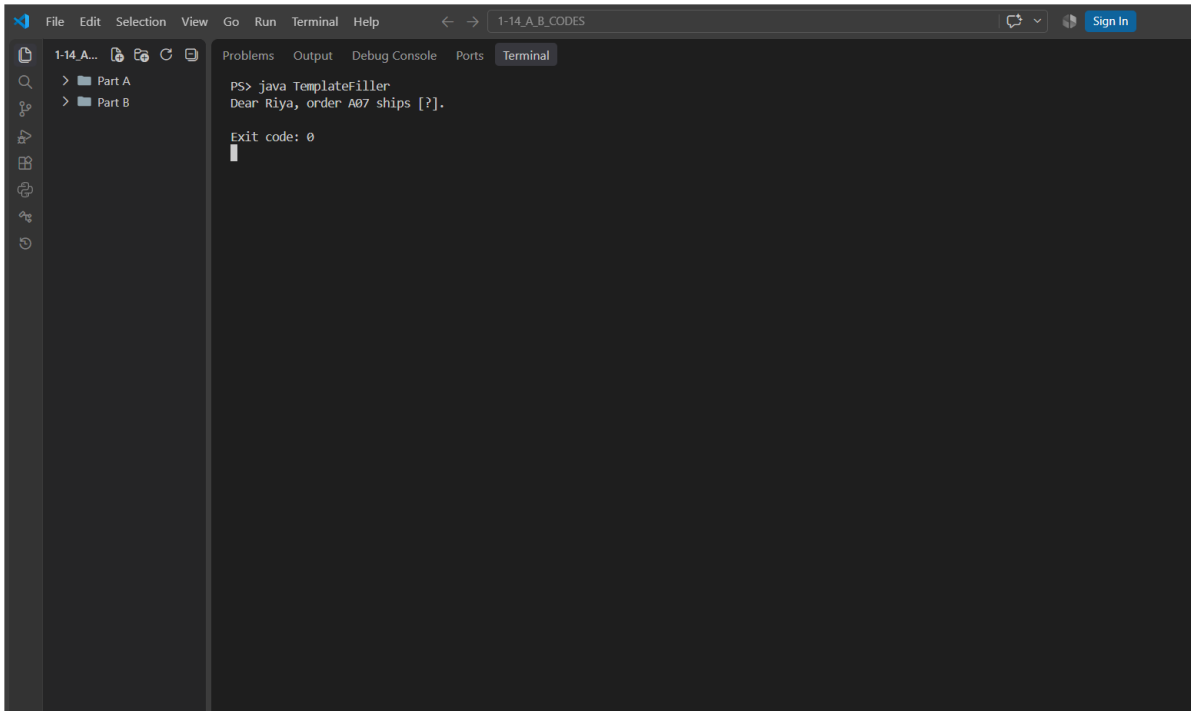
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Validate and parse text using String methods, StringBuilder, regular expressions, and reusable helper classes.

PRACTICAL 5-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 5 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Shape Areas, Payroll, Media Late Fees and Part B: Abstract Account subclasses and polymorphism.

CODE SCREENSHOT:



Code Screenshot 1

Payroll.java

Part A > Practical 5-A > Payroll.java > ...

```java
abstract class Employee {
    String name, id;

    Employee(String n, String i) {
        name = n;
        id = i;
    }

    abstract double monthlySalary();
}

class FullTime extends Employee {
    double salary;

    FullTime(String n, String i, double s) {
        super(n, i);
        salary = s;
    }

    double monthlySalary() {
        return salary;
    }
}

class PartTime extends Employee {
    int hours;
    double rate;

    PartTime(String n, String i, int h, double r) {
        super(n, i);
        hours = h;
        rate = r;
    }

    double monthlySalary() {
        return hours * rate;
    }
}

class Intern extends Employee {
    double stipend;

    Intern(String n, String i, double s) {
        super(n, i);
        stipend = s;
    }

    double monthlySalary() {
```

Code Screenshot 2

ShapeDemo.java

Part A > Practical 5-A > ShapeDemo.java > ...

```java
abstract class Shape {
    abstract double area();
}

class Circle extends Shape {
    double r;

    Circle(double r) {
        this.r = r;
    }

    double area() {
        return Math.PI * r * r;
    }
}

class Rectangle extends Shape {
    double w, h;

    Rectangle(double w, double h) {
        this.w = w;
        this.h = h;
    }

    double area() {
        return w * h;
    }
}

class Triangle extends Shape {
    double b, h;

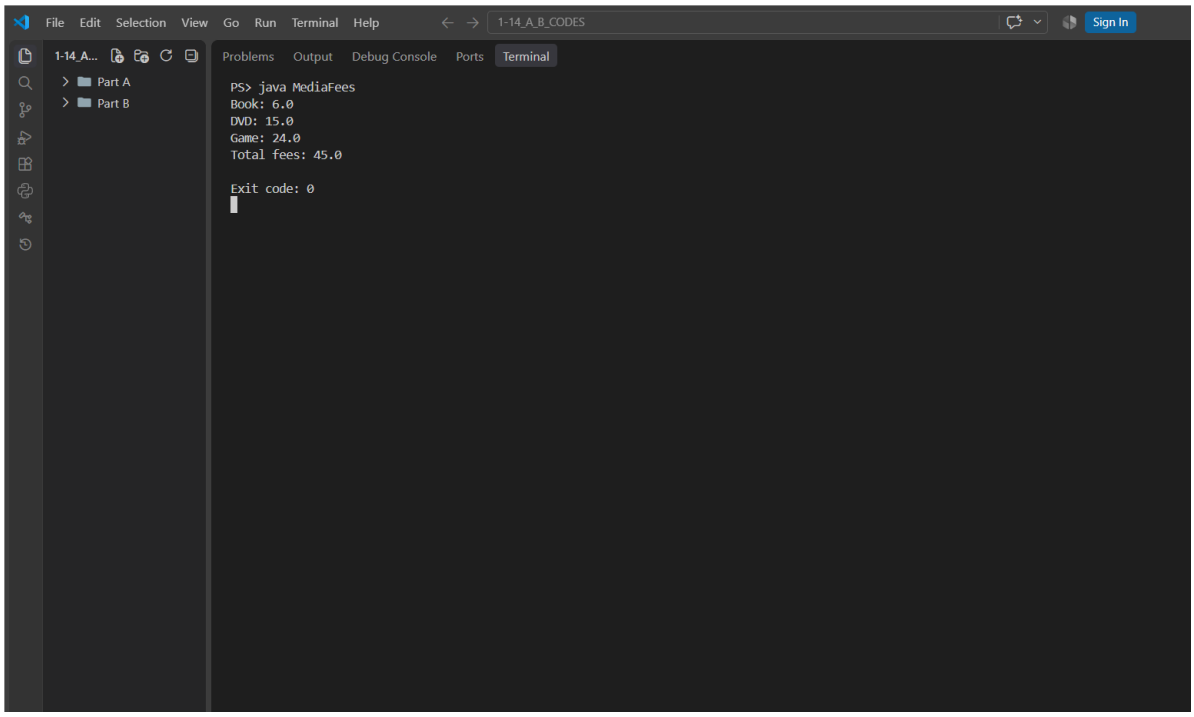
    Triangle(double b, double h) {
        this.b = b;
        this.h = h;
    }

    double area() {
        return b * h / 2;
    }
}

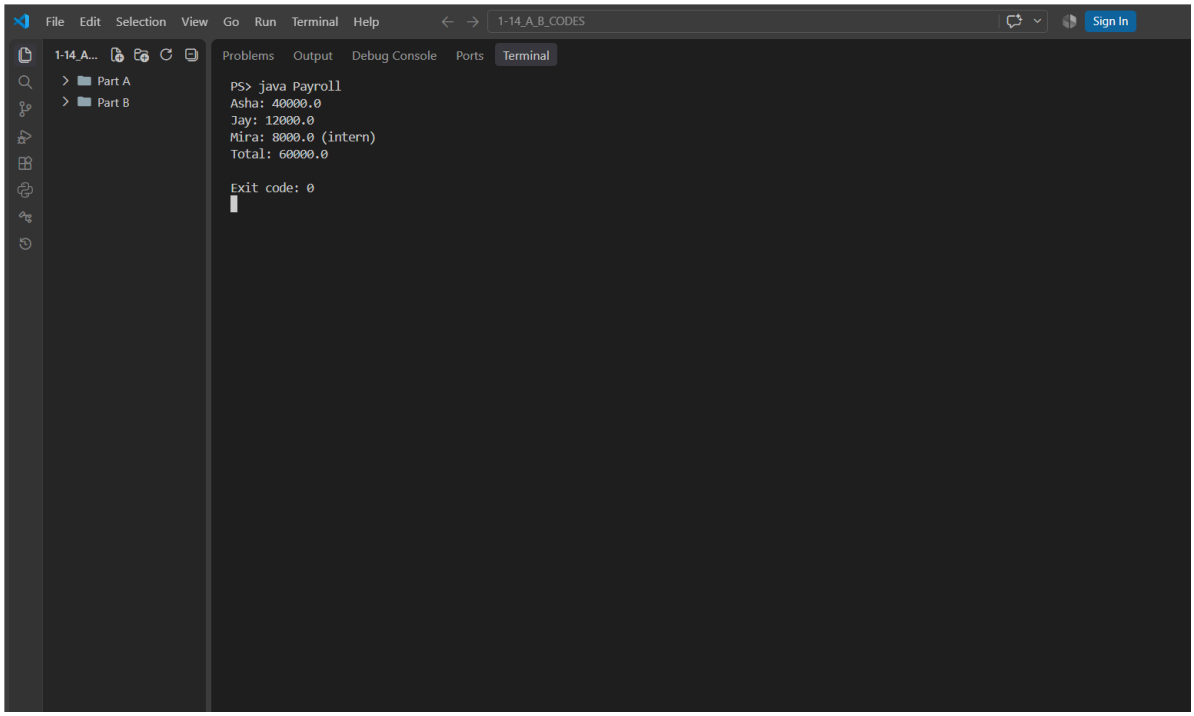
public class ShapeDemo {
    Run | Debug
    public static void main(String[] args) {
        Shape[] shapes = { new Circle(r: 2), new Rectangle(w: 3, h: 4), new Triangle(b: 3, h: 5) };
        double total = 0, largest = 0;
        for (Shape s : shapes) {
```

Code Screenshot 3

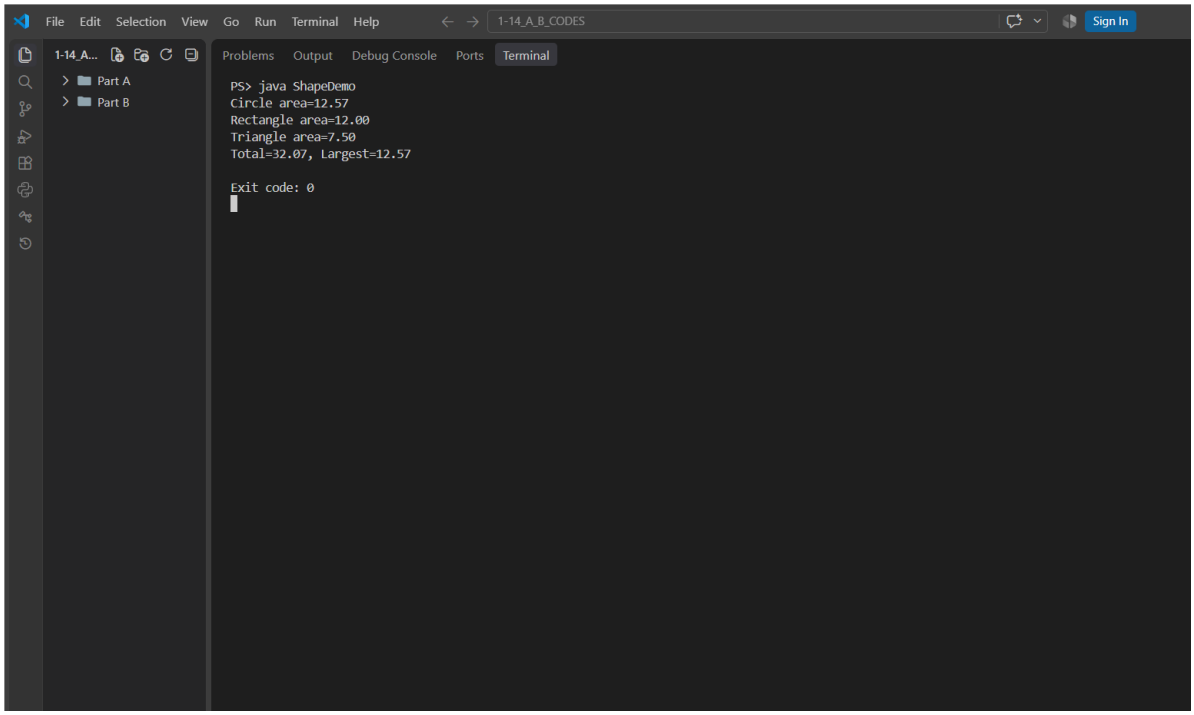
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Use inheritance, abstract classes, method overriding, and runtime polymorphism to model related types.

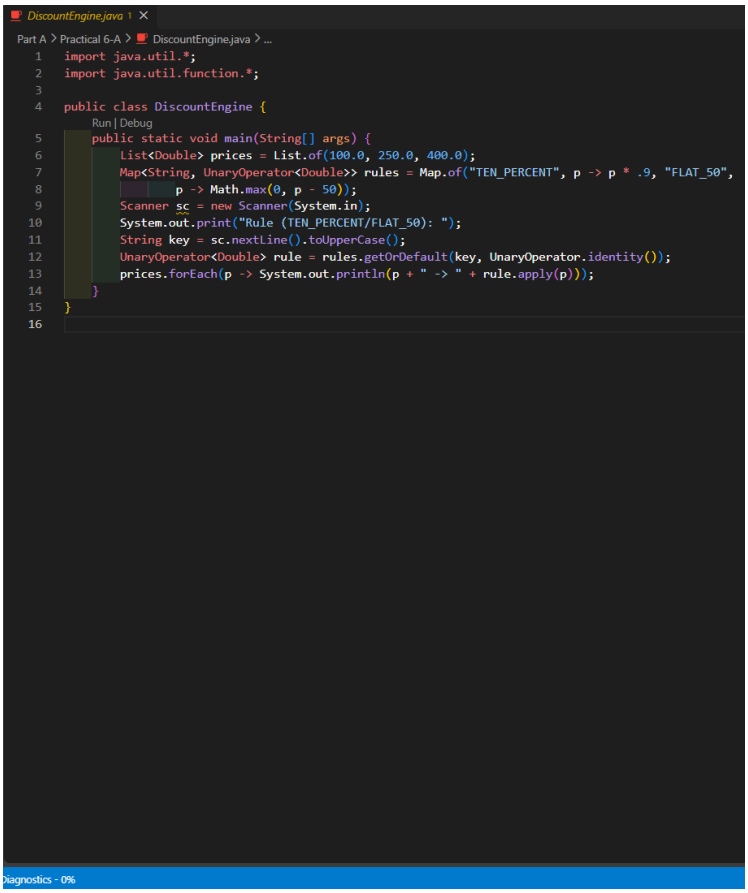
PRACTICAL 6-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 6 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Remote Control, Notification Senders, Discount Engine and Part B: Interfaces, lambda rule, packages and JAR concepts.

CODE SCREENSHOT:



Code Screenshot 1



```java
@FunctionalInterface
interface Notifier {
    void send(String message);
}

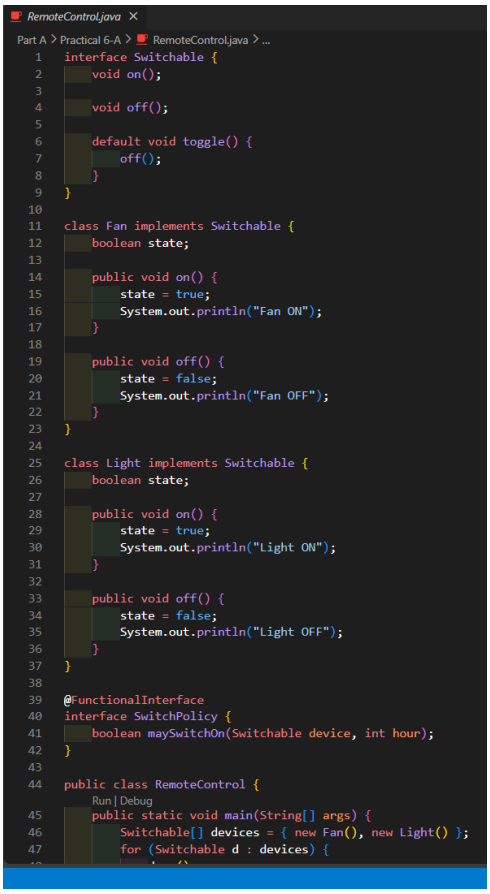
interface Urgent {
}

class EmailNotifier implements Notifier {
    public void send(String m) {
        System.out.println("Email: " + m);
    }
}

class SmsNotifier implements Notifier, Urgent {
    public void send(String m) {
        System.out.println("SMS: " + m);
    }
}

public class NotificationSenders {
    Run | Debug
    public static void main(String[] args) {
        Notifier[] senders = { new EmailNotifier(), new SmsNotifier() };
        for (Notifier n : senders) {
            n.send(message: "System maintenance at 8 PM");
            if (n instanceof Urgent)
                n.send(message: "URGENT: System maintenance at 8 PM");
        }
    }
}
```

Code Screenshot 2



```java
interface Switchable {
    void on();

    void off();

    default void toggle() {
        off();
    }
}

class Fan implements Switchable {
    boolean state;

    public void on() {
        state = true;
        System.out.println("Fan ON");
    }

    public void off() {
        state = false;
        System.out.println("Fan OFF");
    }
}

class Light implements Switchable {
    boolean state;

    public void on() {
        state = true;
        System.out.println("Light ON");
    }

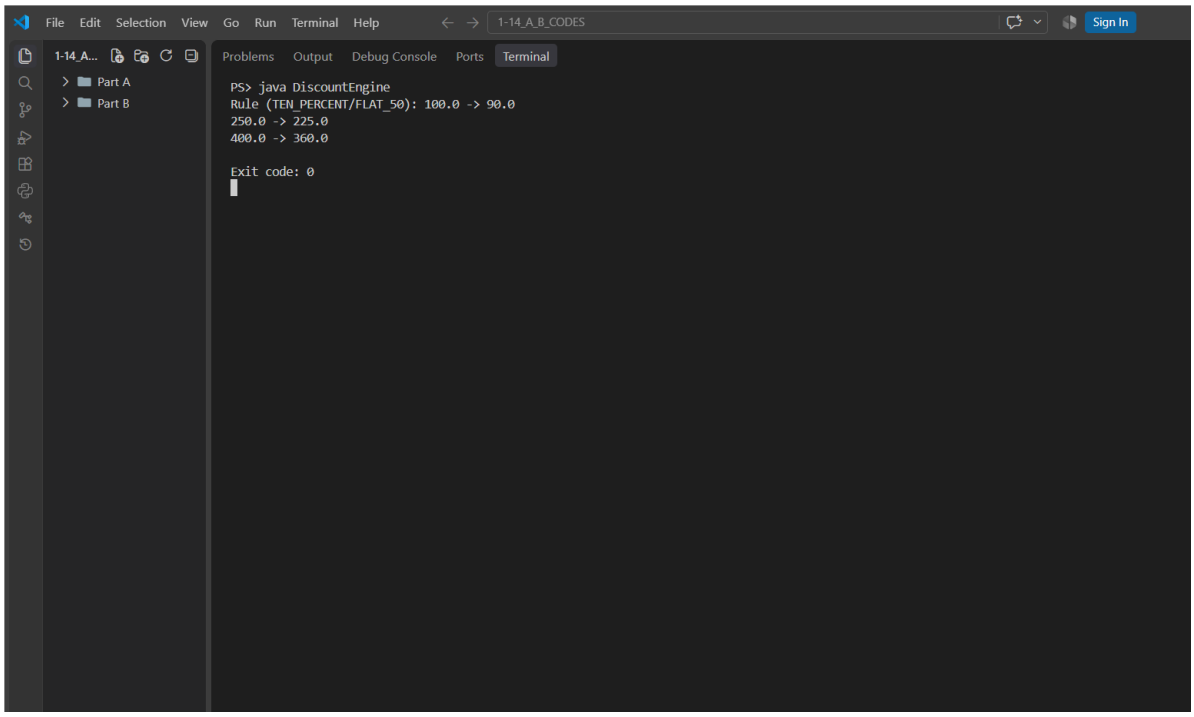
    public void off() {
        state = false;
        System.out.println("Light OFF");
    }
}

@FunctionalInterface
interface SwitchPolicy {
    boolean maySwitchOn(Switchable device, int hour);
}

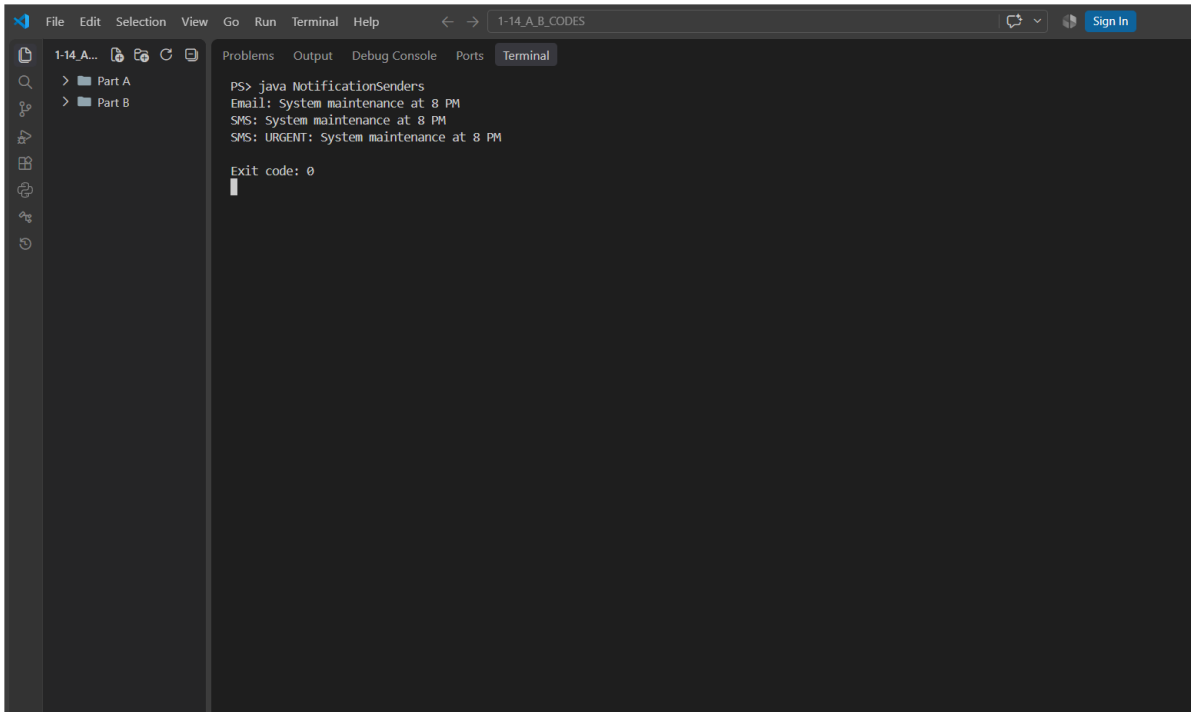
public class RemoteControl {
    Run | Debug
    public static void main(String[] args) {
        Switchable[] devices = { new Fan(), new Light() };
        for (Switchable d : devices) {
```

Code Screenshot 3

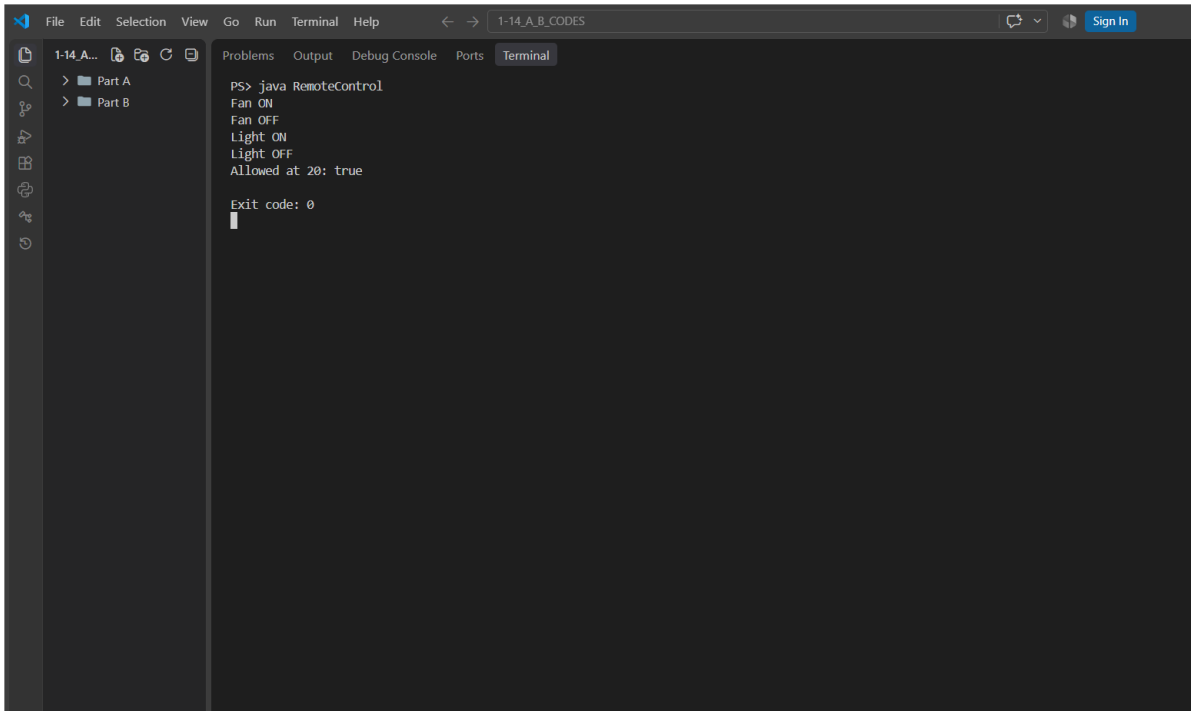
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Design programs with interfaces, functional interfaces, lambda expressions, and default methods.

PRACTICAL 7-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 7 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Annotation Form Validator, Mini Test Runner, Column Mapper and Part B: Custom annotations and reflection validator.

CODE SCREENSHOT:

ColumnMapper.java

Part A > Practical 7-A > ColumnMapper.java > ...

```java
import java.lang.annotation.*;
import java.lang.reflect.*;

@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.FIELD)
@interface Column {
    String name();
}

class Person {
    @Column(name = "name")
    String name;
    @Column(name = "age")
    int age;

    public String toString() {
        return name + " (" + age + ")";
    }
}

public class ColumnMapper {
    Run | Debug
    public static void main(String[] args) throws Exception {
        String[] header = { "age", "name" }, row = { "20", "Anshul" };
        Person p = new Person();
        for (Field f : Person.class.getDeclaredFields()) {
            Column c = f.getAnnotation(Column.class);
            for (int i = 0; i < header.length; i++)
                if (c.name().equals(header[i])) {
                    f.setAccessible(true);
                    if (f.getType() == int.class)
                        f.setInt(p, Integer.parseInt(row[i]));
                    else
                        f.set(p, row[i]);
                }
        }
        System.out.println(p);
    }
}

```

Code Screenshot 1

FormValidator.java

Part A > Practical 7-A > FormValidator.java > ...

```java
import java.lang.annotation.*;
import java.lang.reflect.*;
import java.util.*;

@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.FIELD)
@interface NotBlank {
}

@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.FIELD)
@interface MaxLength {
    int value();
}

class SignupForm {
    @NotBlank
    String name = " ";
    @MaxLength(10)
    String username = "verylongusername";

    SignupForm(String n, String u) {
        name = n;
        username = u;
    }
}

public class FormValidator {
    Run | Debug
    public static void main(String[] args) throws Exception {
        SignupForm form = new SignupForm(n: " ", u: "verylongusername");
        List<String> errors = new ArrayList<>();
        for (Field f : form.getClass().getDeclaredFields()) {
            f.setAccessible(true);
            Object v = f.get(form);
            if (f.isAnnotationPresent(NotBlank.class) && (!(v instanceof String) || ((String) v).isBlank()))
                errors.add(f.getName() + " must not be blank");
            MaxLength max = f.getAnnotation(MaxLength.class);
            if (max != null && v instanceof String s && s.length() > max.value())
                errors.add(f.getName() + " exceeds " + max.value());
        }
        System.out.println(errors);
    }
}

```

Code Screenshot 2

MiniTestRunner.java

Part A > Practical 7-A > MiniTestRunner.java > ...

```java
import java.lang.annotation.*;
import java.lang.reflect.*;

@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.METHOD)
@interface Run {
}

class DemoTests {
    @Run
    void testOne() {
        System.out.println("testOne passed");
    }

    @Run
    void testTwo() {
        System.out.println("testTwo passed");
    }

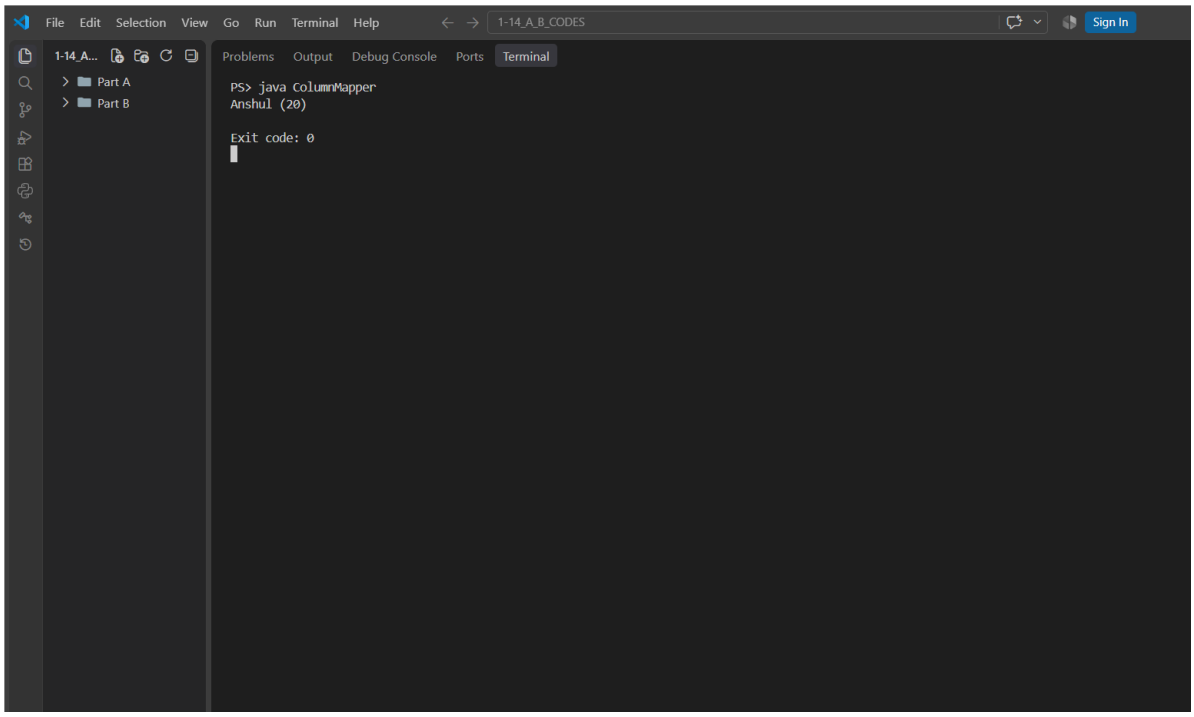
    void helper() {
        System.out.println("ignored");
    }
}

public class MiniTestRunner {
    Run | Debug
    public static void main(String[] args) throws Exception {
        Object test = new DemoTests();
        int count = 0;
        for (Method m : test.getClass().getDeclaredMethods())
            if (m.isAnnotationPresent(Run.class)) {
                m.invoke(test);
                count++;
            }
        System.out.println("Tests run: " + count);
    }
}

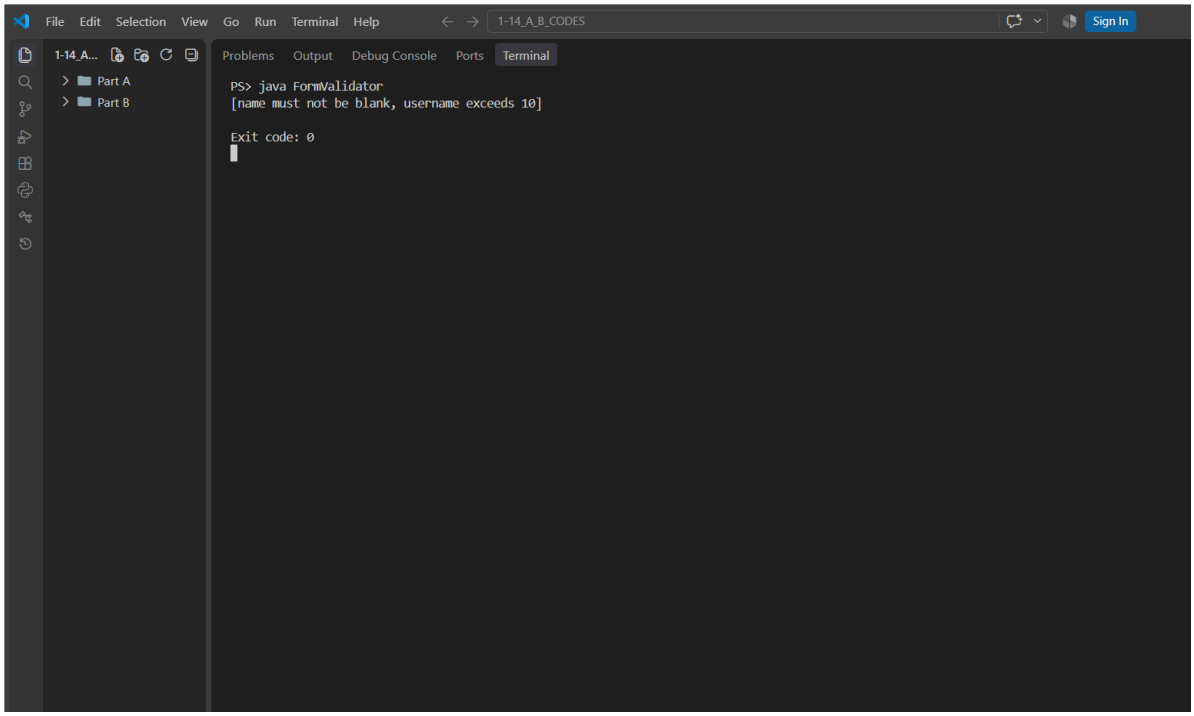
```

Code Screenshot 3

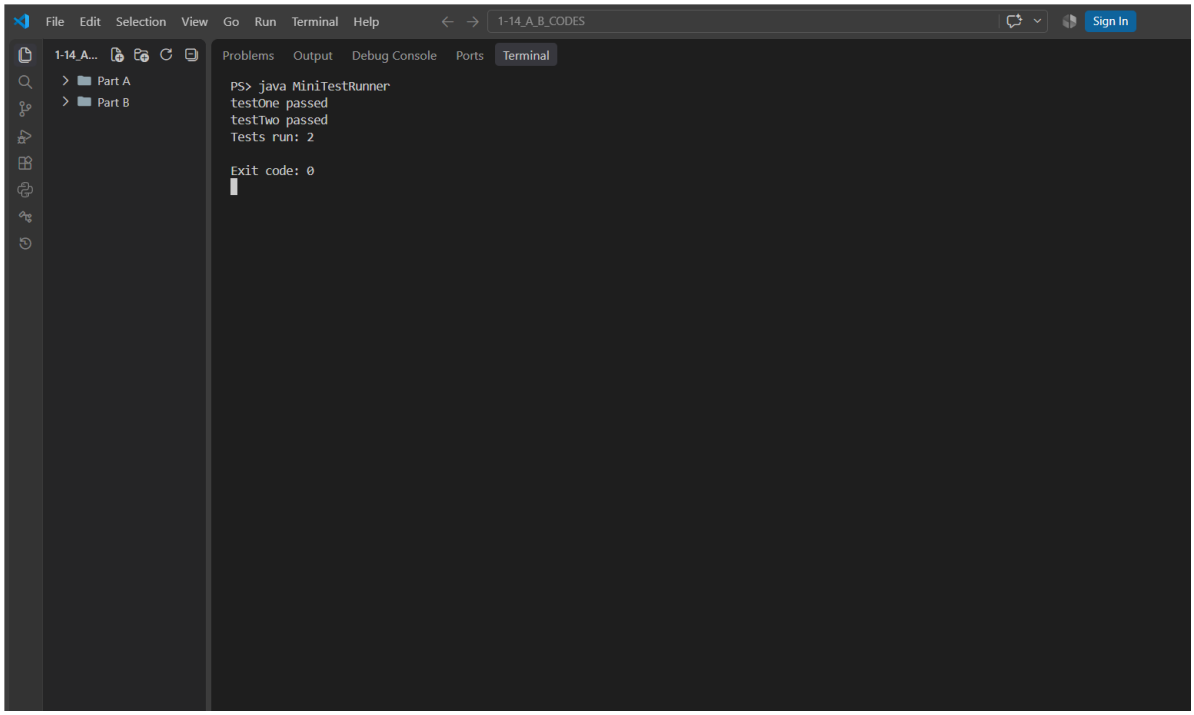
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Create runtime annotations and inspect object fields through reflection for validation.

PRACTICAL 8-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 8 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Guarded Calculator, Warehouse Exceptions, AutoCloseable and Part B: Bank exception hierarchy and transfer handling.

CODE SCREENSHOT:

GuardedCalculator.java 1

Part A > Practical 8-A > GuardedCalculator.java > ...

```java
import java.util.*;

class DivideByZeroException extends Exception {
    DivideByZeroException(String m) {
        super(m);
    }
}

public class GuardedCalculator {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        while (true) {
            try {
                System.out.print("First number (q to quit): ");
                String x = sc.next();
                if (x.equalsIgnoreCase("q"))
                    break;
                double a = Double.parseDouble(x);
                System.out.print("Operator and second number: ");
                String op = sc.next();
                double b = Double.parseDouble(sc.next());
                double result;
                switch (op) {
                    case "+" -> result = a + b;
                    case "-" -> result = a - b;
                    case "*" -> result = a * b;
                    case "/" -> {
                        if (b == 0)
                            throw new DivideByZeroException(m: "Cannot divide by zero");
                        result = a / b;
                    }
                    default -> throw new IllegalArgumentException("Unknown operator");
                }
                System.out.println("Result: " + result);
                break;
            } catch (NumberFormatException e) {
                System.out.println("Invalid number");
            } catch (DivideByZeroException | IllegalArgumentException e) {
                System.out.println(e.getMessage());
            } finally {
                System.out.println("Attempt logged");
            }
        }
    }
}
```

Code Screenshot 1

ResourceDemo.java

Part A > Practical 8-A > ResourceDemo.java > ...

```java
class SampleResource implements AutoCloseable {
    public void use() {
        System.out.println("Using resource");
    }

    public void close() {
        System.out.println("Resource closed");
    }
}

public class ResourceDemo {
    Run | Debug
    public static void main(String[] args) {
        try (SampleResource r = new SampleResource()) {
            r.use();
            if (args.length > 0)
                throw new RuntimeException("Demo failure");
        } catch (RuntimeException e) {
            System.out.println("Reported: " + e.getMessage());
        }
    }
}

```

Code Screenshot 2

Warehouse.java

Part A > Practical 8-A > Warehouse.java > ...

```java
import java.util.*;

class OutOfStockException extends Exception {
    final int shortfall;

    OutOfStockException(int s) {
        super("Out of stock by " + s);
        shortfall = s;
    }
}

class InvalidQuantityException extends Exception {
    InvalidQuantityException() {
        super("Quantity must be positive");
    }
}

public class Warehouse {
    private final Map<String, Integer> stock = new HashMap<>(Map.of("pen", 5, "book", 2));

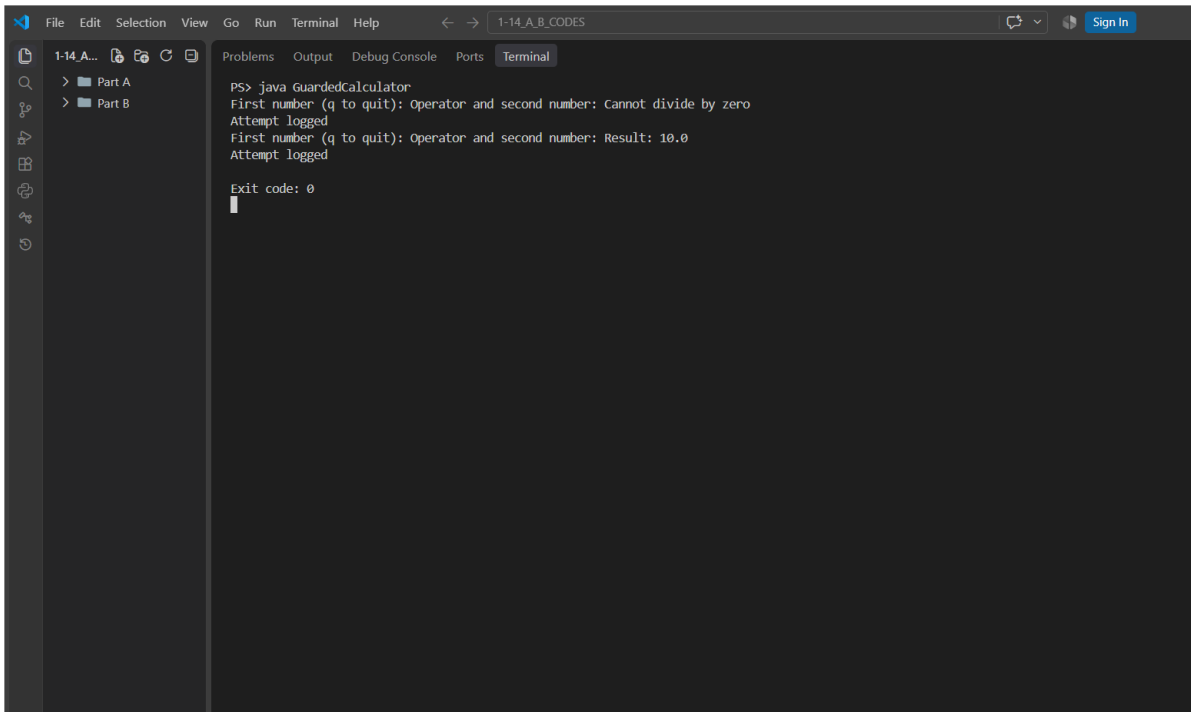
    public void issue(String item, int qty) throws OutOfStockException, InvalidQuantityException {
        if (qty <= 0)
            throw new InvalidQuantityException();
        int available = stock.getOrDefault(item, 0);
        if (qty > available)
            throw new OutOfStockException(qty - available);
        stock.put(item, available - qty);
        System.out.println("Issued " + qty + " " + item);
    }

    Run | Debug
    public static void main(String[] args) {
        Warehouse w = new Warehouse();
        Object[][] req = { { "pen", 2 }, { "book", 5 }, { "pen", 0 } };
        for (Object[] r : req)
            try {
                w.issue((String) r[0], (int) r[1]);
            } catch (Exception e) {
                System.out.println(e.getMessage());
            }
    }
}

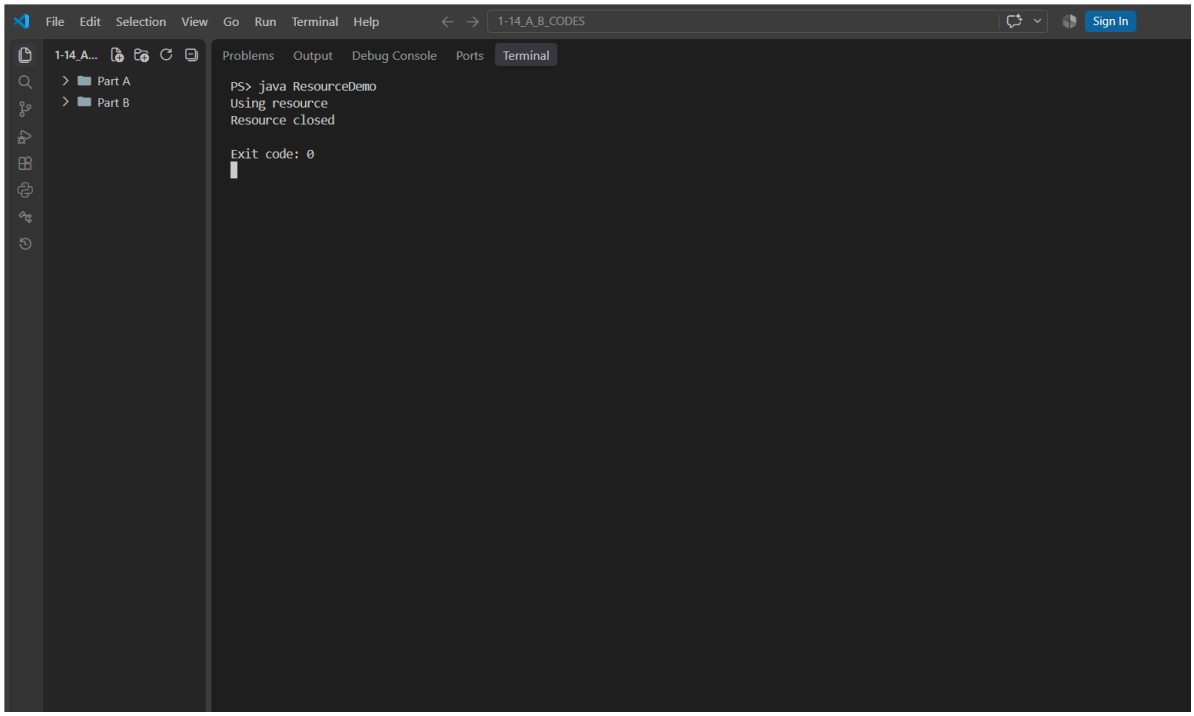
```

Code Screenshot 3

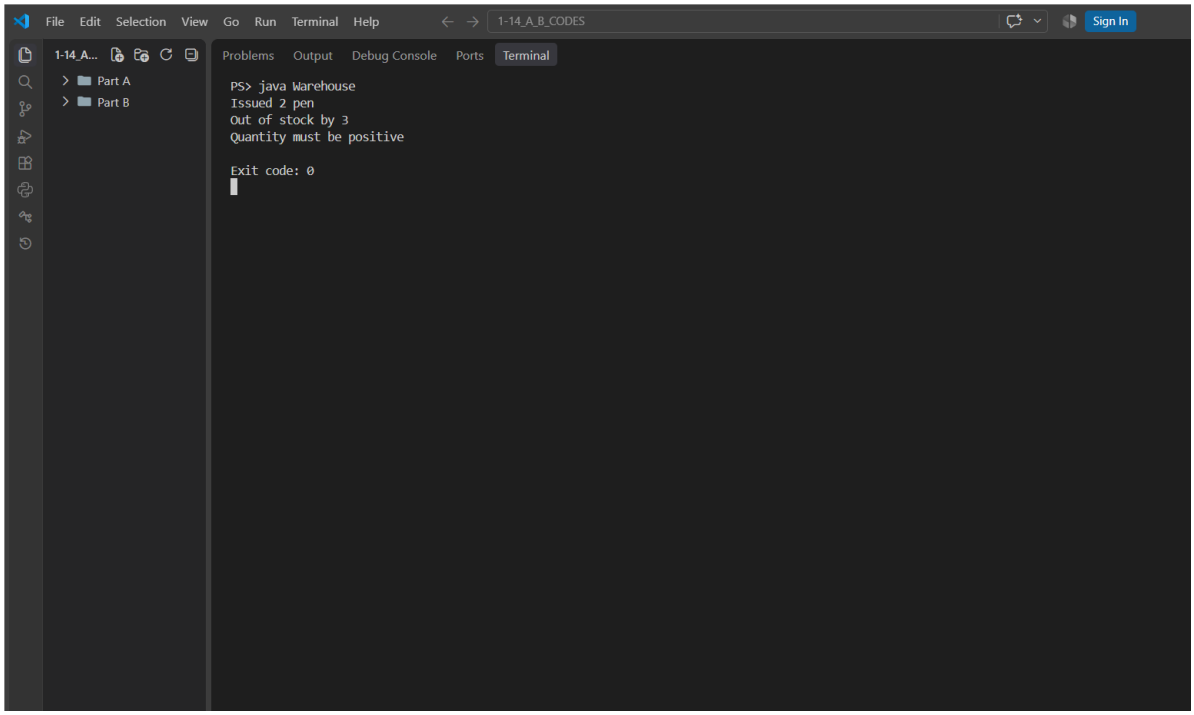
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Handle invalid operations safely with custom exceptions, try-catch-finally, and try-with-resources.

PRACTICAL 9-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 9 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Counter Race, Seat Booking Race, Parallel Sum and Part B: Runnable workers and synchronized account operations.

CODE SCREENSHOT:

CounterRace.java

Part A > Practical 9-A > CounterRace.java > ...

```java
public class CounterRace {
    static class Counter {
        int value = 0;

        synchronized void increment() {
            value++;
        }
    }

    Run | Debug
    public static void main(String[] args) throws InterruptedException {
        Counter c = new Counter();
        Thread[] ts = new Thread[10];
        for (int i = 0; i < ts.length; i++) {
            ts[i] = new Thread(() -> {
                for (int j = 0; j < 1000; j++)
                    c.increment();
            });
            ts[i].start();
        }
        for (Thread t : ts)
            t.join();
        System.out.println("Final counter: " + c.value + " (expected 10000)");
    }
}

```

Code Screenshot 1

ParallelSum.java

Part A > Practical 9-A > ParallelSum.java > ...

```java
import java.util.*;

public class ParallelSum {
    Run | Debug
    public static void main(String[] args) throws InterruptedException {
        int[] values = new int[10000];
        Arrays.fill(values, 1);
        long[] sums = new long[4];
        Thread[] ts = new Thread[4];
        for (int i = 0; i < 4; i++) {
            final int k = i;
            ts[i] = new Thread(() -> {
                for (int j = k * 2500; j < (k + 1) * 2500; j++)
                    sums[k] += values[j];
            });
            ts[i].start();
        }
        for (Thread t : ts)
            t.join();
        System.out.println("Parallel sum: " + Arrays.stream(sums).sum());
    }
}

```

Code Screenshot 2

SeatBooking.java

Part A > Practical 9-A > SeatBooking.java > ...

```java
public class SeatBooking {
    static class Cinema {
        int seatsLeft = 5;

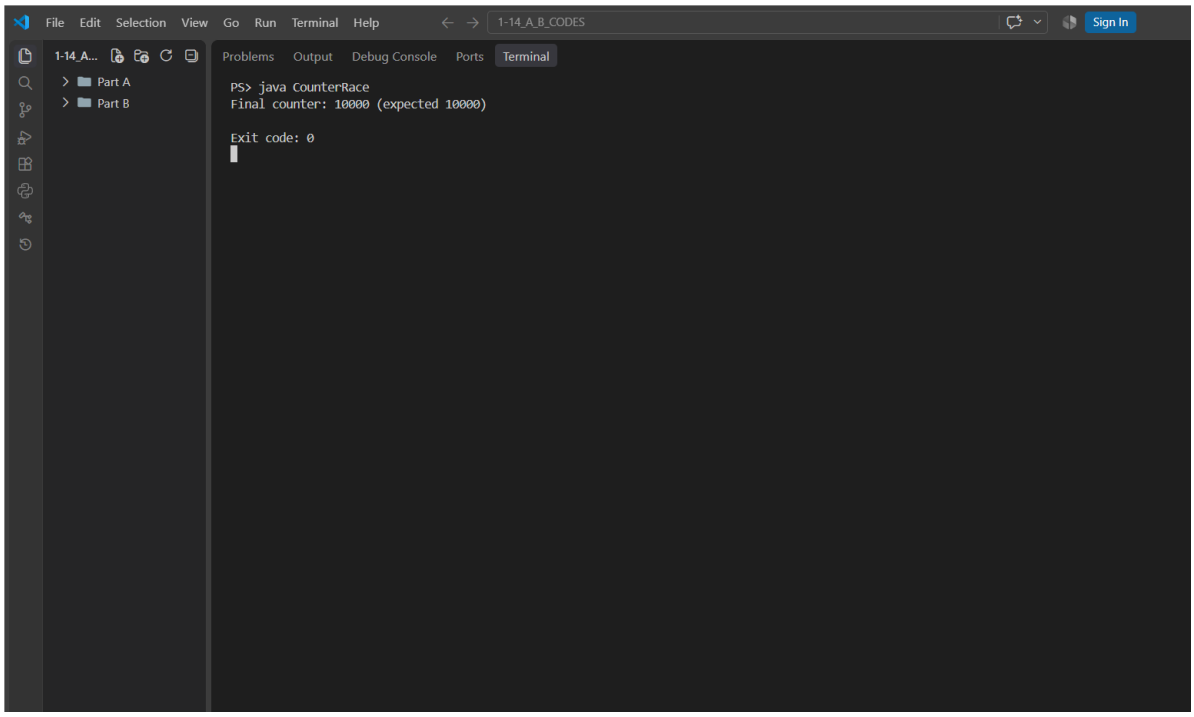
        synchronized boolean book() {
            if (seatsLeft > 0) {
                seatsLeft--;
                return true;
            }
            return false;
        }
    }

    Run | Debug
    public static void main(String[] args) throws InterruptedException {
        Cinema c = new Cinema();
        int[] booked = { 0 };
        Thread[] ts = new Thread[10];
        for (int i = 0; i < 10; i++) {
            ts[i] = new Thread(() -> {
                if (c.book())
                    synchronized (booked) {
                        booked[0]++;
                    }
            });
            ts[i].start();
        }
        for (Thread t : ts)
            t.join();
        System.out.println("Booked: " + booked[0] + ", seats left: " + c.seatsLeft);
    }
}

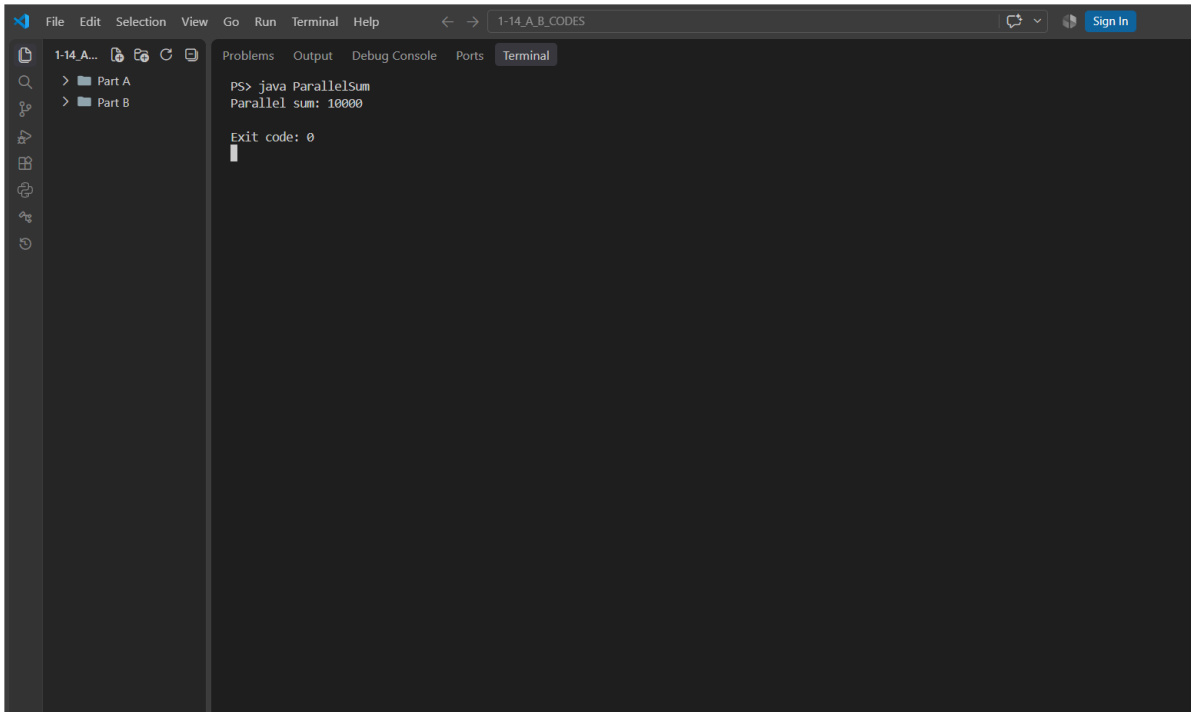
```

Code Screenshot 3

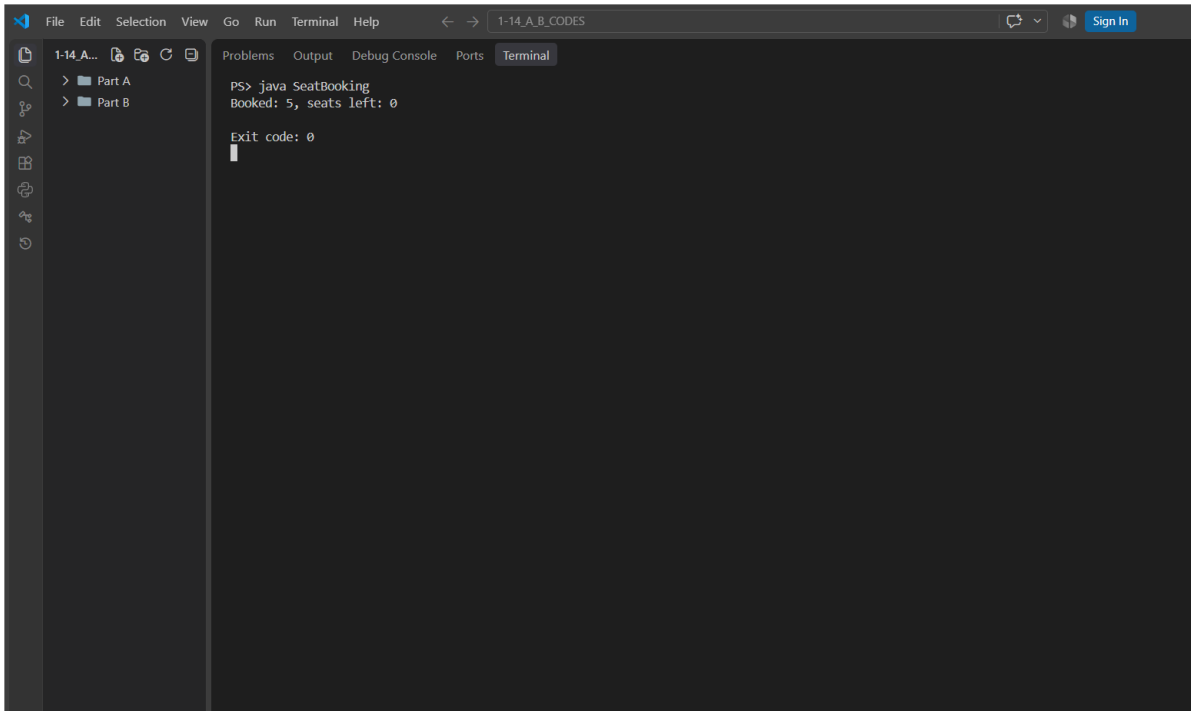
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Explain race conditions and protect shared mutable state with synchronization and thread coordination.

PRACTICAL 10-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 10 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Thread Pool, Producer-Consumer, Deadlock Demo and Part B: TransactionProcessor and batch operations.

CODE SCREENSHOT:

DeadlockDemo.java

Part A > Practical 10-A > DeadlockDemo.java > DeadlockDemo > main(String[])

```java
public class DeadlockDemo {
    Run | Debug
    public static void main(String[] args) {
        Object a = new Object(), b = new Object();
        Thread t1 = new Thread(() -> {
            synchronized (a) {
                sleep();
                synchronized (b) {
                    System.out.println("t1 acquired both");
                }
            }
        });
        Thread t2 = new Thread(() -> {
            synchronized (b) {
                sleep();
                synchronized (a) {
                    System.out.println("t2 acquired both");
                }
            }
        });
        t1.start();
        t2.start();
        System.out.println("Demo may deadlock; stop manually if both threads block.");
    }

    static void sleep() {
        try {
            Thread.sleep(100);
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }
    }
}
```

Code Screenshot 1

ProducerConsumer.java

Part A > Practical 10-A > ProducerConsumer.java > ...

```java
import java.util.concurrent.*;

public class ProducerConsumer {
    Run | Debug
    public static void main(String[] args) throws Exception {
        BlockingQueue<Integer> buffer = new ArrayBlockingQueue<>(3);
        Thread producer = new Thread(() -> {
            try {
                for (int i = 1; i <= 5; i++) {
                    buffer.put(i);
                    System.out.println("Produced " + i);
                }
                buffer.put(-1);
            } catch (InterruptedException e) {
                Thread.currentThread().interrupt();
            }
        });
        Thread consumer = new Thread(() -> {
            try {
                while (true) {
                    int n = buffer.take();
                    if (n == -1)
                        break;
                    System.out.println("Consumed " + n);
                }
            } catch (InterruptedException e) {
                Thread.currentThread().interrupt();
            }
        });
        producer.start();
        consumer.start();
        producer.join();
        consumer.join();
    }
}

```

Code Screenshot 2

ThreadPoolRunner.java

Part A > Practical 10-A > ThreadPoolRunner.java > ...

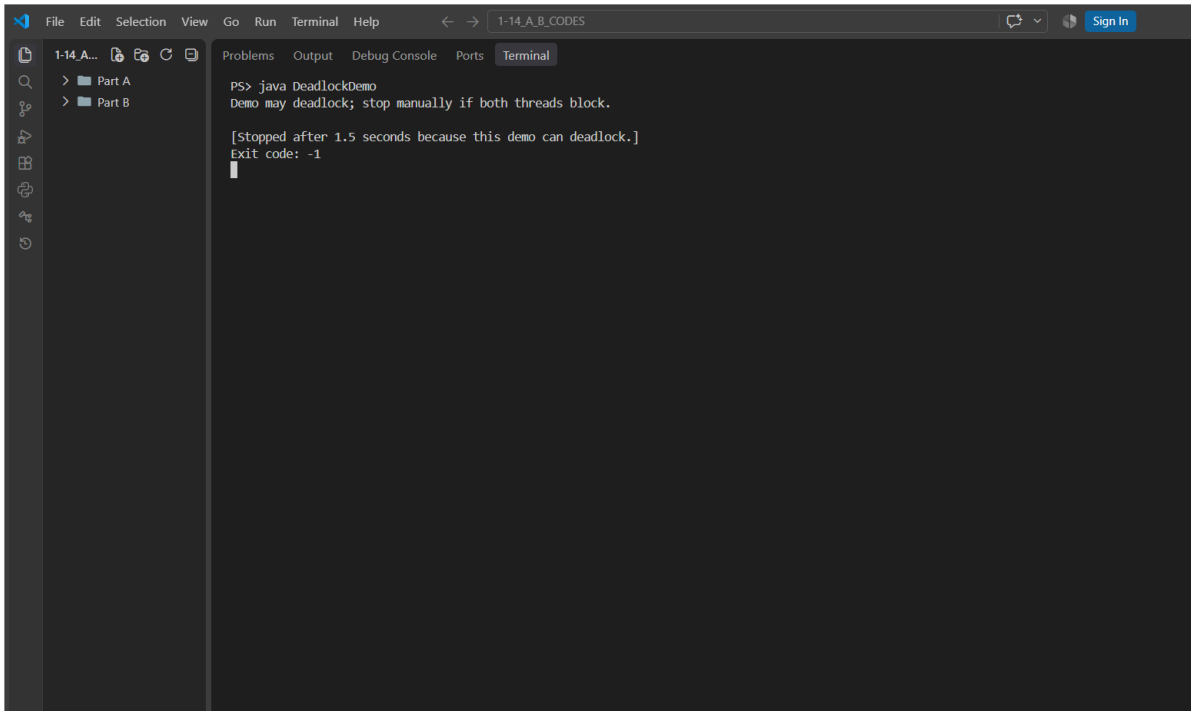
```java
import java.util.concurrent.*;

public class ThreadPoolRunner {
    Run | Debug
    public static void main(String[] args) throws InterruptedException {
        ExecutorService pool = Executors.newFixedThreadPool(3);
        for (int i = 1; i <= 10; i++) {
            final int n = i;
            pool.submit(() -> System.out.println("Task " + n + " on " + Thread.currentThread().getName()));
        }
        pool.shutdown();
        pool.awaitTermination(1, TimeUnit.MINUTES);
    }
}

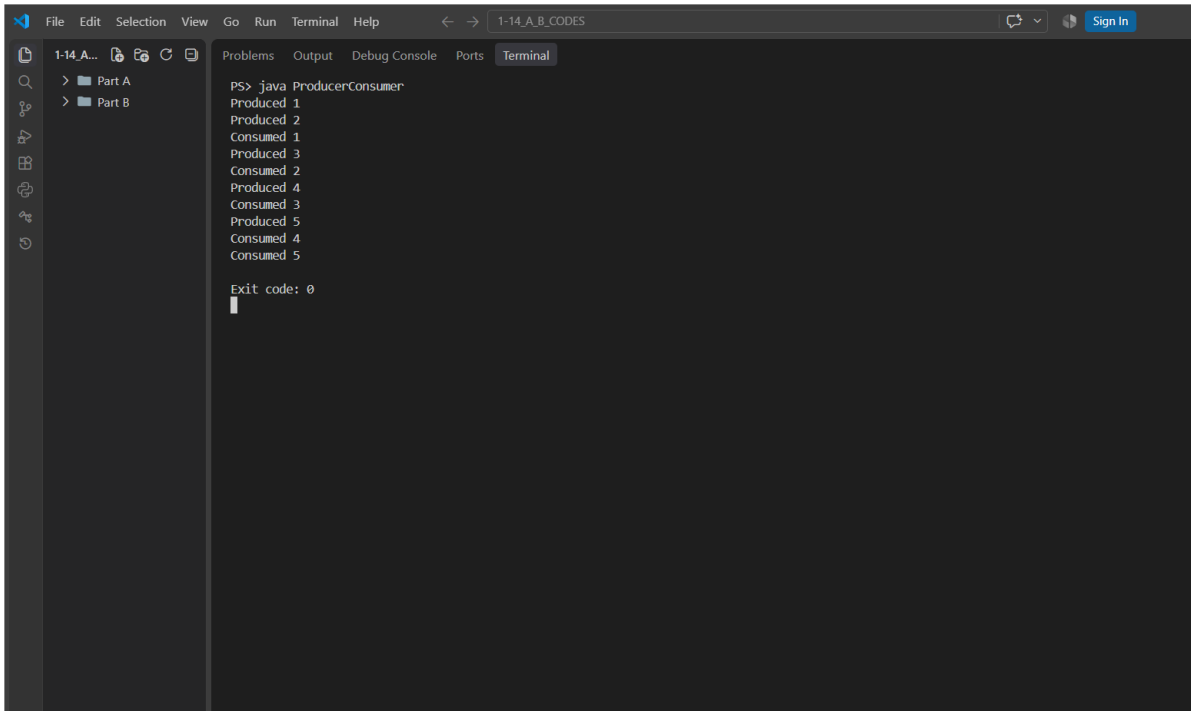
```

Code Screenshot 3

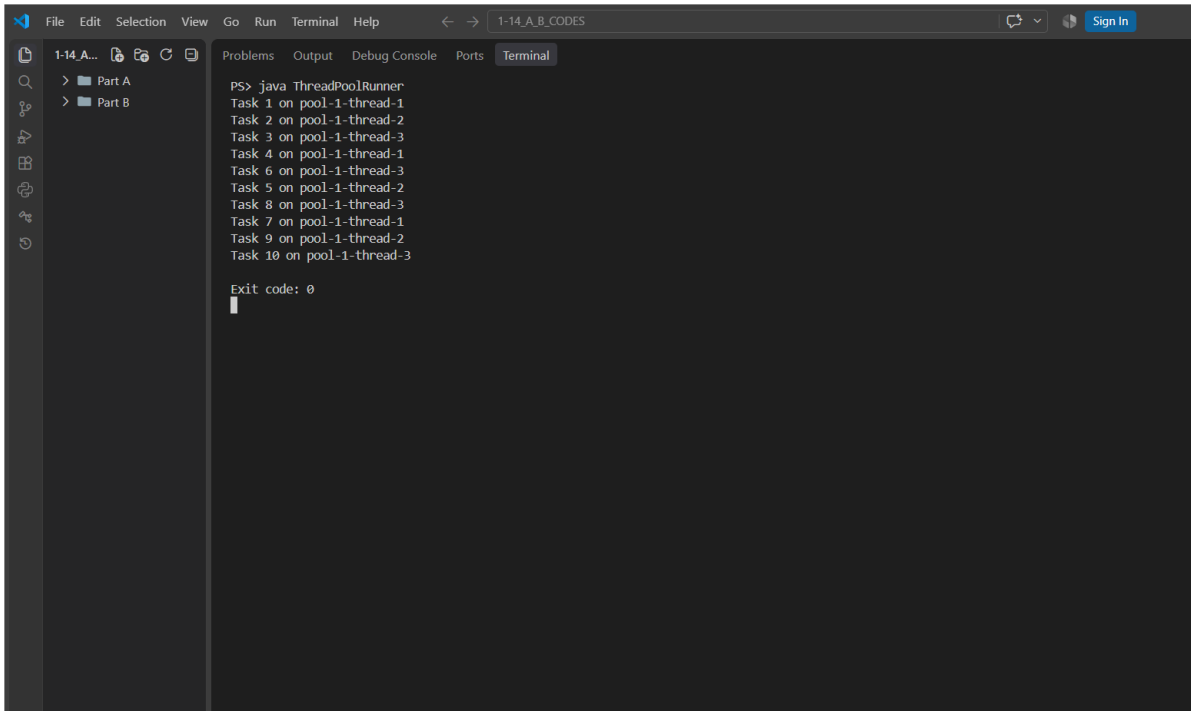
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Use executor services and producer-consumer coordination to manage concurrent tasks.

PRACTICAL 11-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 11 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: File Persistence and Report and Part B: Load/save account data and report.

CODE SCREENSHOT:

AccountFileStore.java 1

Part A > Practical 11-A > AccountFileStore.java > ...

```java
import java.io.*;
import java.nio.file.*;
import java.util.*;

public class AccountFileStore {
    Run | Debug
    public static void main(String[] args) throws Exception {
        Path p = Path.of("accounts.csv");
        List<String> rows = List.of("AC0001,Anshul,5000", "AC0002,Riya,2500");
        Files.write(p, rows);
        List<String> loaded = Files.readAllLines(p);
        System.out.println("Persistence verified: " + rows.equals(loaded));
        loaded.forEach(System.out::println);
    }
}

```

Code Screenshot 1

BankReport.java

Part A > Practical 11-A > BankReport.java > ...

```java
import java.util.*;

public class BankReport {
    record Tx(String type, long amount) {
    }

    Run | Debug
    public static void main(String[] args) {
        List<Tx> tx = List.of(new Tx(type: "DEPOSIT", amount: 5000), new Tx(type: "WITHDRAW", amount: 1200), new Tx(type: "DEPOSIT", amount: 300));
        long deposits = tx.stream().filter(t -> t.type.equals("DEPOSIT")).mapToLong(Tx::amount).sum();
        long withdrawals = tx.stream().filter(t -> t.type.equals("WITHDRAW")).mapToLong(Tx::amount).sum();
        System.out.println("Transactions: " + tx.size());
        System.out.println("Deposits: " + deposits);
        System.out.println("Withdrawals: " + withdrawals);
        System.out.println("Net: " + (deposits - withdrawals));
    }
}

```

Code Screenshot 2

PersistenceDemo.java

Part A > Practical 11-A > PersistenceDemo.java > ...

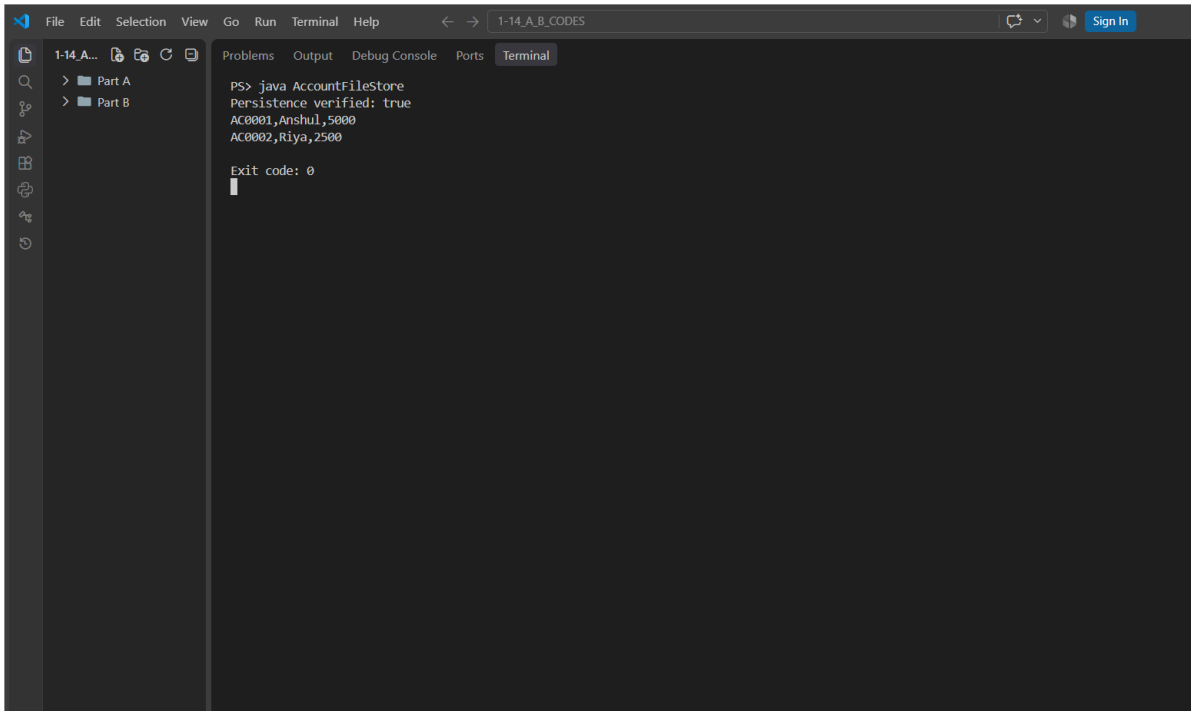
```java
import java.io.*;
import java.nio.file.*;
import java.util.*;

public class PersistenceDemo {
    Run | Debug
    public static void main(String[] args) throws IOException {
        Path path = Path.of("accounts.txt");
        List<String> accounts = Files.exists(path) ? Files.readAllLines(path) : new ArrayList<>();
        if (accounts.isEmpty())
            accounts.add("AC0001,Anshul,5000");
        Files.write(path, accounts);
        System.out.println("Loaded accounts:");
        accounts.forEach(System.out::println);
        System.out.println("Saved to " + path.toAbsolutePath());
    }
}

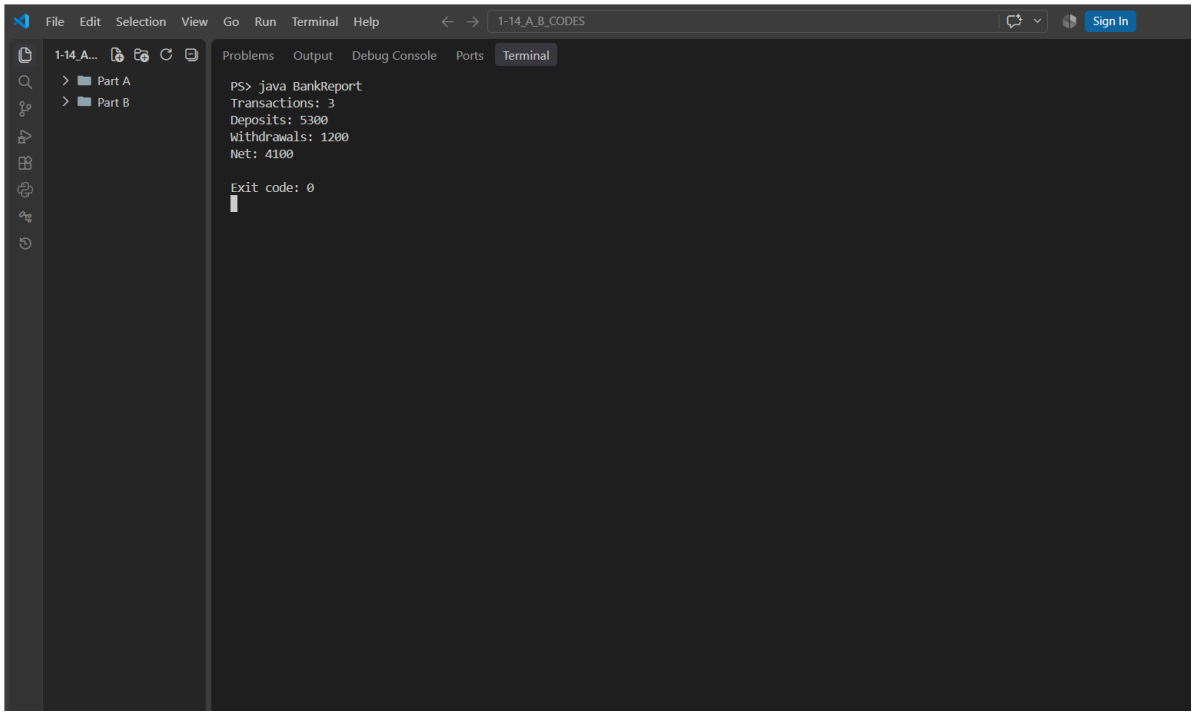
```

Code Screenshot 3

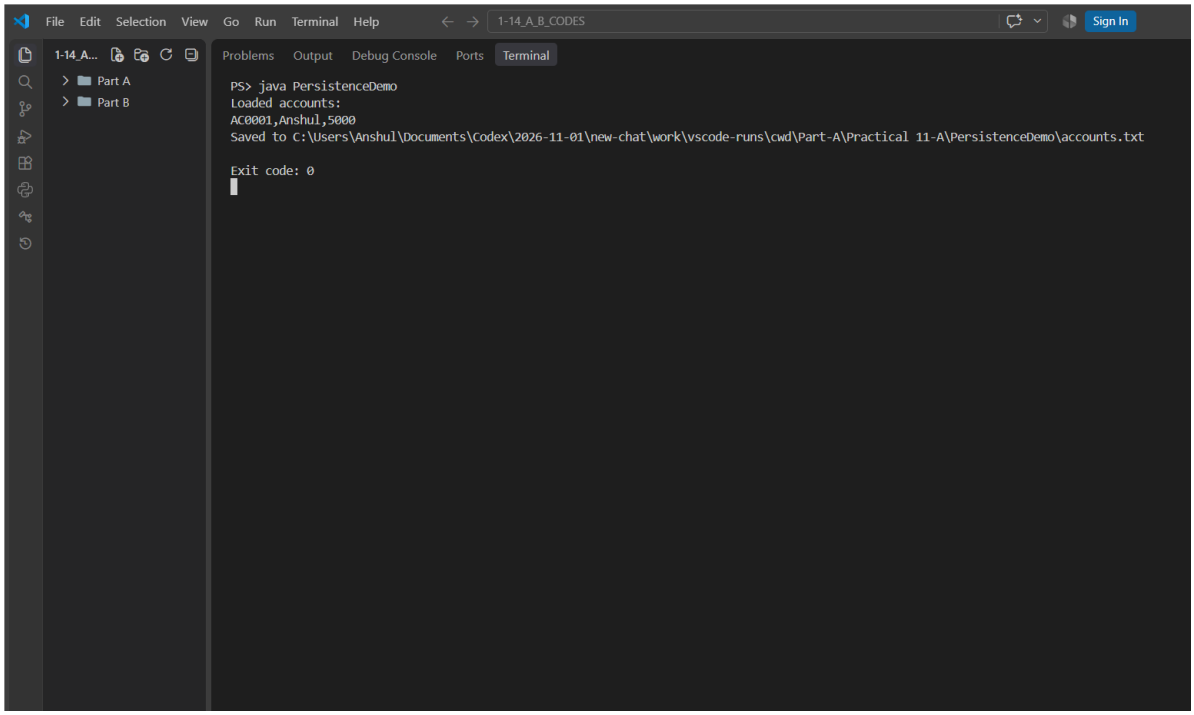
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Persist application data to files, reload saved records, and generate a basic report.

PRACTICAL 12-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 12 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Concurrent Store, Transfer, Thread Lifecycle and Part B: Thread-safe account store.

CODE SCREENSHOT:

ConcurrentStore.java

Part A > Practical 12-A > ConcurrentStore.java > ...

```java
import java.util.concurrent.*;

public class ConcurrentStore {
    Run | Debug
    public static void main(String[] args) throws InterruptedException {
        ConcurrentHashMap<String, Long> balances = new ConcurrentHashMap<>();
        balances.put("AC0001", 0L);
        Thread[] ts = new Thread[10];
        for (int i = 0; i < ts.length; i++) {
            ts[i] = new Thread(() -> {
                for (int j = 0; j < 1000; j++)
                    balances.compute("AC0001", (k, v) -> v + 1);
            });
            ts[i].start();
        }
        for (Thread t : ts)
            t.join();
        System.out.println("Balance: " + balances.get("AC0001"));
    }
}

```

Code Screenshot 1

ConcurrentTransfer.java

Part A > Practical 12-A > ConcurrentTransfer.java > ...

```java
import java.util.concurrent.*;

public class ConcurrentTransfer {
    static class Bank {
        final ConcurrentHashMap<String, Long> b = new ConcurrentHashMap<>();

        synchronized boolean transfer(String from, String to, long amount) {
            if (amount <= 0 || b.get(from) < amount)
                return false;
            b.put(from, b.get(from) - amount);
            b.put(to, b.get(to) + amount);
            return true;
        }
    }

    Run | Debug
    public static void main(String[] args) {
        Bank bank = new Bank();
        bank.b.put("A", 1000L);
        bank.b.put("B", 1000L);
        System.out.println(bank.transfer(from: "A", to: "B", amount: 250));
        System.out.println(bank.b);
    }
}

```

Code Screenshot 2

ThreadLifecycle.java

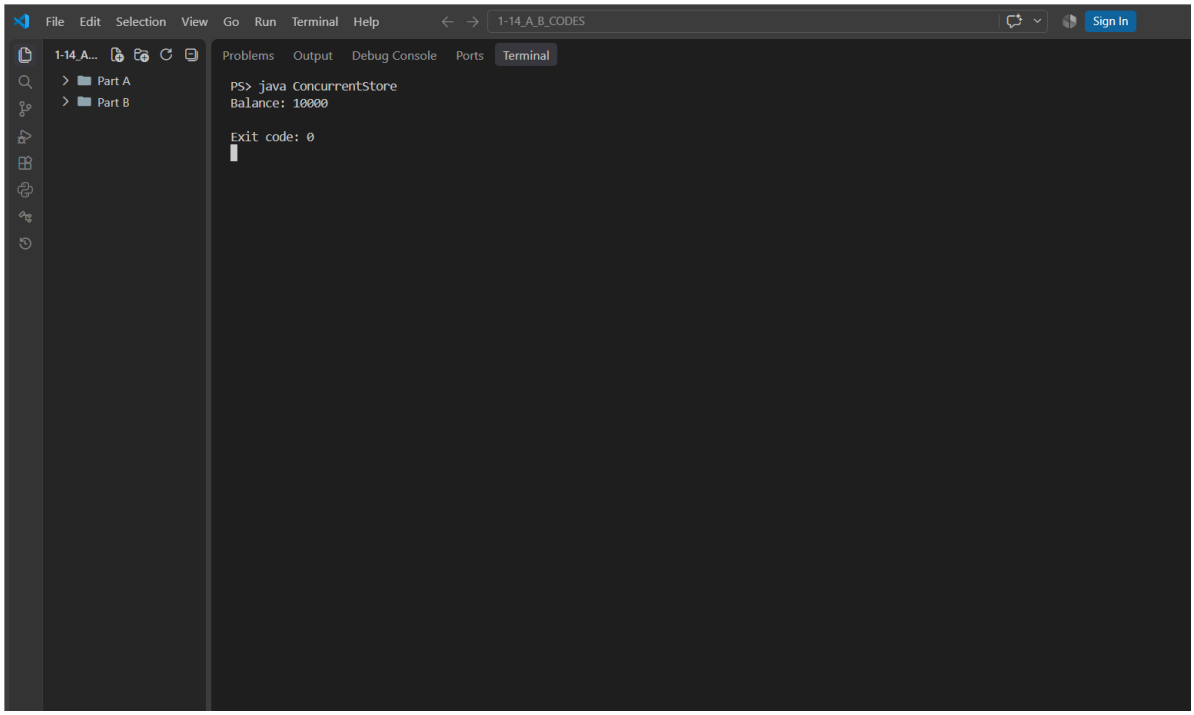
Part A > Practical 12-A > ThreadLifecycle.java > ...

```java
public class ThreadLifecycle {
    Run | Debug
    public static void main(String[] args) throws InterruptedException {
        Thread t = new Thread(() -> System.out.println("Running task"));
        System.out.println(t.getState());
        t.start();
        t.join();
        System.out.println(t.getState());
    }
}

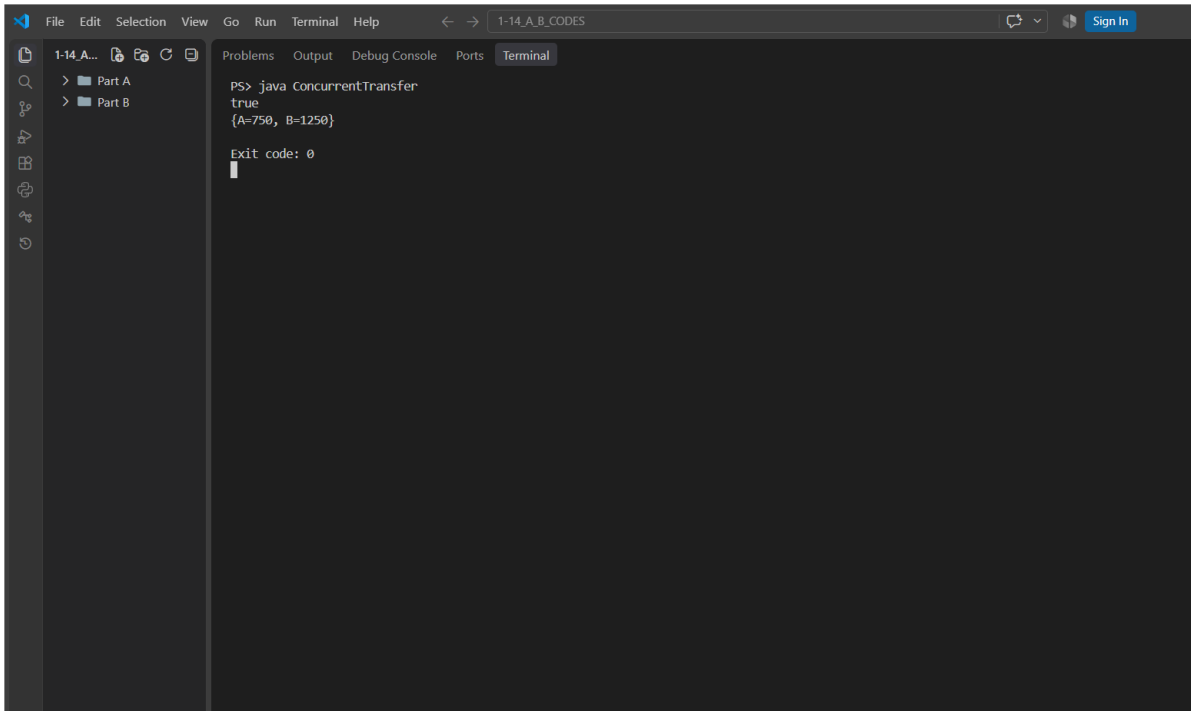
```

Code Screenshot 3

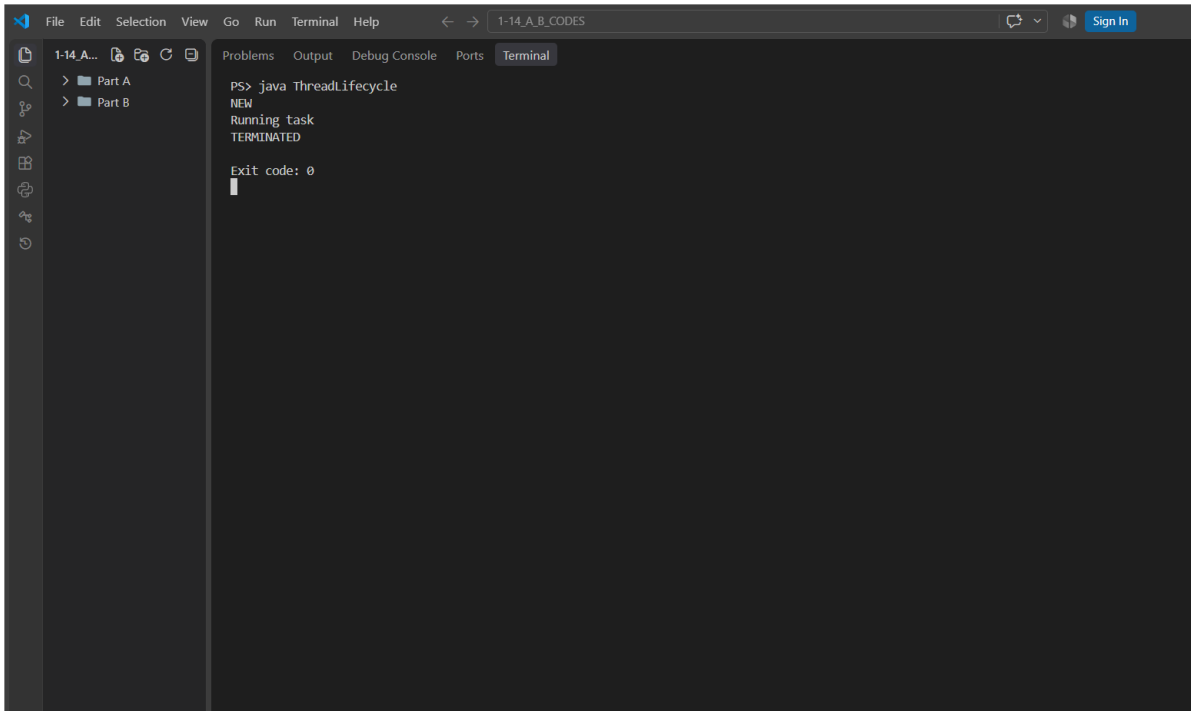
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Use concurrent collections and synchronized operations to maintain consistency under concurrent access.

PRACTICAL 13-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 13 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Generic repository / bounded generics and Part B: Generic Repository<T> using @Id reflection.

CODE SCREENSHOT:

GenericRepository.java

Part A > Practical 13-A > GenericRepository.java > ...

```java
import java.lang.annotation.*;
import java.lang.reflect.*;
import java.util.*;

@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.FIELD)
@interface Id {
}

class Account {
    @Id
    private final String accountNumber;
    private final String owner;
    private long balance;

    Account(String n, String o, long b) {
        accountNumber = n;
        owner = o;
        balance = b;
    }

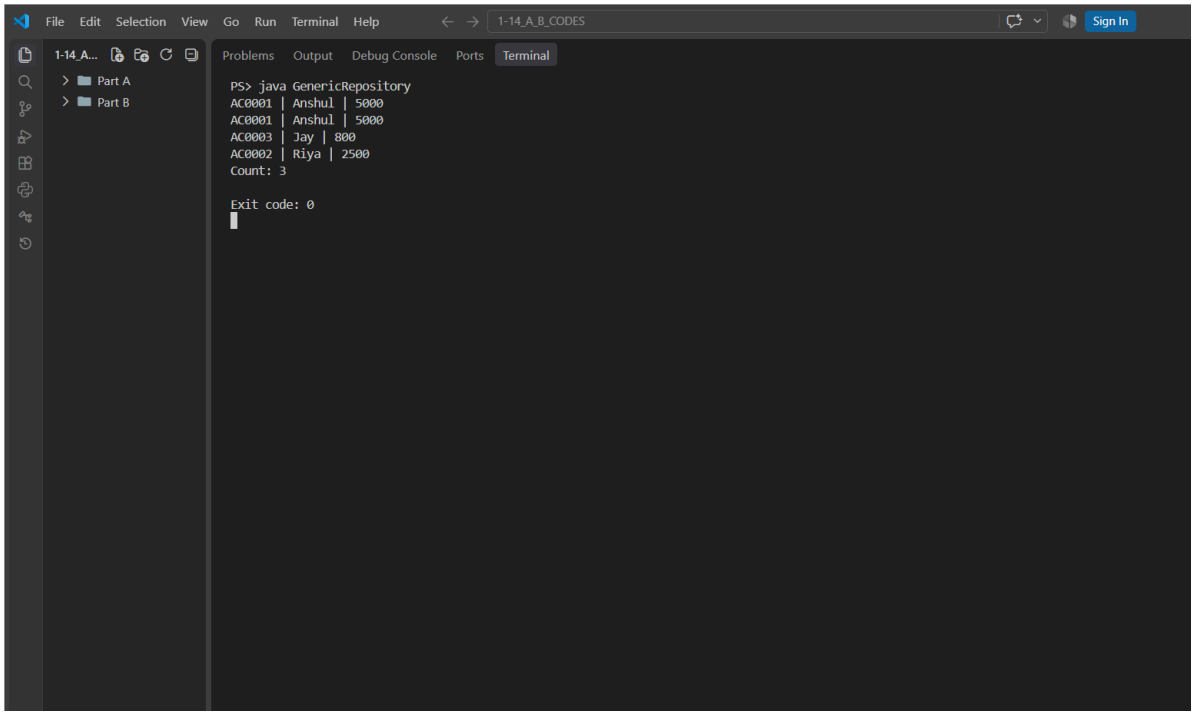
    public String toString() {
        return accountNumber + " | " + owner + " | " + balance;
    }
}

class Repository<T> {
    private final Map<String, T> data = new HashMap<>();

    public T save(T entity) {
        try {
            for (Field f : entity.getClass().getDeclaredFields())
                if (f.isAnnotationPresent(Id.class)) {
                    f.setAccessible(true);
                    data.put(String.valueOf(f.get(entity)), entity);
                    return entity;
                }
            throw new IllegalArgumentException("No @Id field");
        } catch (IllegalAccessException e) {
            throw new RuntimeException(e);
        }
    }

    @SafeVarargs
    public final void saveAll(T... items) {
        for (T item : items)
            save(item);
    }
```

OUTPUT SCREENSHOT:



LEARNING OUTCOMES:

Build reusable type-safe generic components and access entity identifiers using reflection.

PRACTICAL 14-A

Object Oriented Programming

AIM:

To assemble and test a small layered Java application by creating a model, a service that manages data using an ArrayList, and a menu-driven main class; then write JUnit tests to verify successful additions and invalid-action handling.

CODE SCREENSHOT:

LayeredApp.java 1

14-A > LayeredApp.java > { } app

```java
package app;

import service.TaskService;
import java.util.Scanner;

public class LayeredApp {
    public static void main(String[] args) {
        TaskService service = new TaskService();
        try (Scanner sc = new Scanner(System.in)) {
            while (true) {
                System.out.println("\n1. Add task  2. List tasks  0. Exit");
                System.out.print("Choice: ");
                String choice = sc.nextLine().trim();
                try {
                    switch (choice) {
                        case "1" -> {
                            System.out.print("Task title: ");
                            service.performAction("ADD", sc.nextLine());
                            System.out.println("Task added.");
                        }
                        case "2" -> service.performAction("LIST", "");
                        case "0" -> {
                            System.out.println("Goodbye.");
                            return;
                        }
                        default -> service.performAction(choice, "");
                    }
                } catch (IllegalArgumentException ex) {
                    System.out.println(ex.getMessage());
                }
            }
        }
    }
}

```

Code Screenshot 1

Task.java 1 ×

14-A > Task.java > Task

```java
package model;

public record Task(int id, String title) {
    public Task {
        if (title == null || title.isBlank()) {
            throw new IllegalArgumentException("Task title cannot be blank");
        }
    }
}

```

Code Screenshot 2

TaskService.java 1 ×

14-A > TaskService.java > ...

```java
package service;

import model.Task;
import java.util.ArrayList;
import java.util.List;

public class TaskService {
    private final List<Task> tasks = new ArrayList<>();
    private int nextId = 1;

    public Task addTask(String title) {
        Task task = new Task(nextId++, title);
        tasks.add(task);
        return task;
    }

    public int count() {
        return tasks.size();
    }

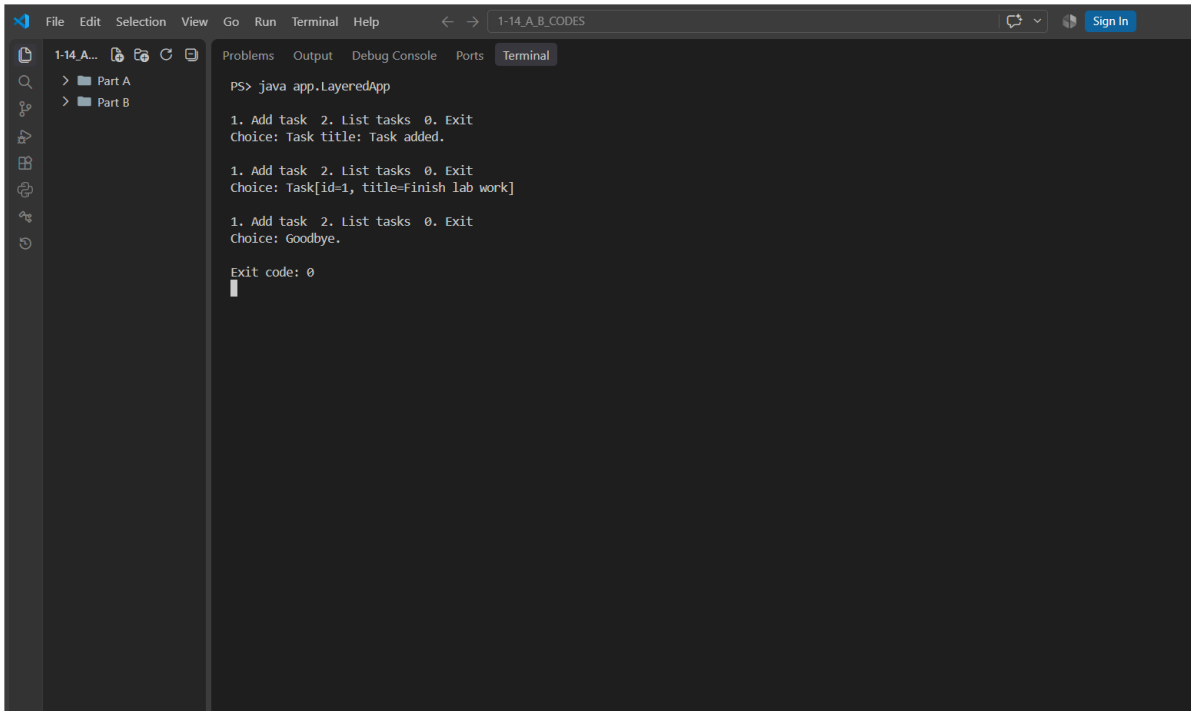
    public List<Task> getAll() {
        return List.copyOf(tasks);
    }

    public void performAction(String action, String value) {
        if ("ADD".equalsIgnoreCase(action)) {
            addTask(value);
        } else if ("LIST".equalsIgnoreCase(action)) {
            getAll().forEach(System.out::println);
        } else {
            throw new IllegalArgumentException("Invalid action: " + action);
        }
    }
}

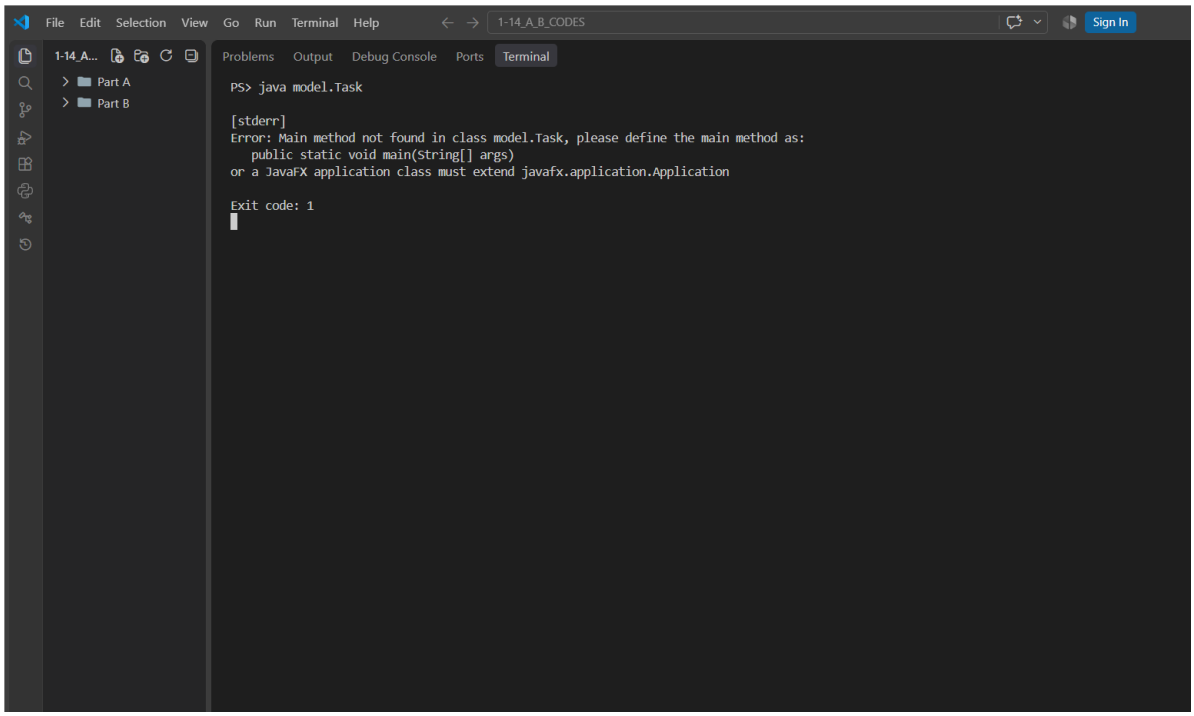
```

Code Screenshot 3

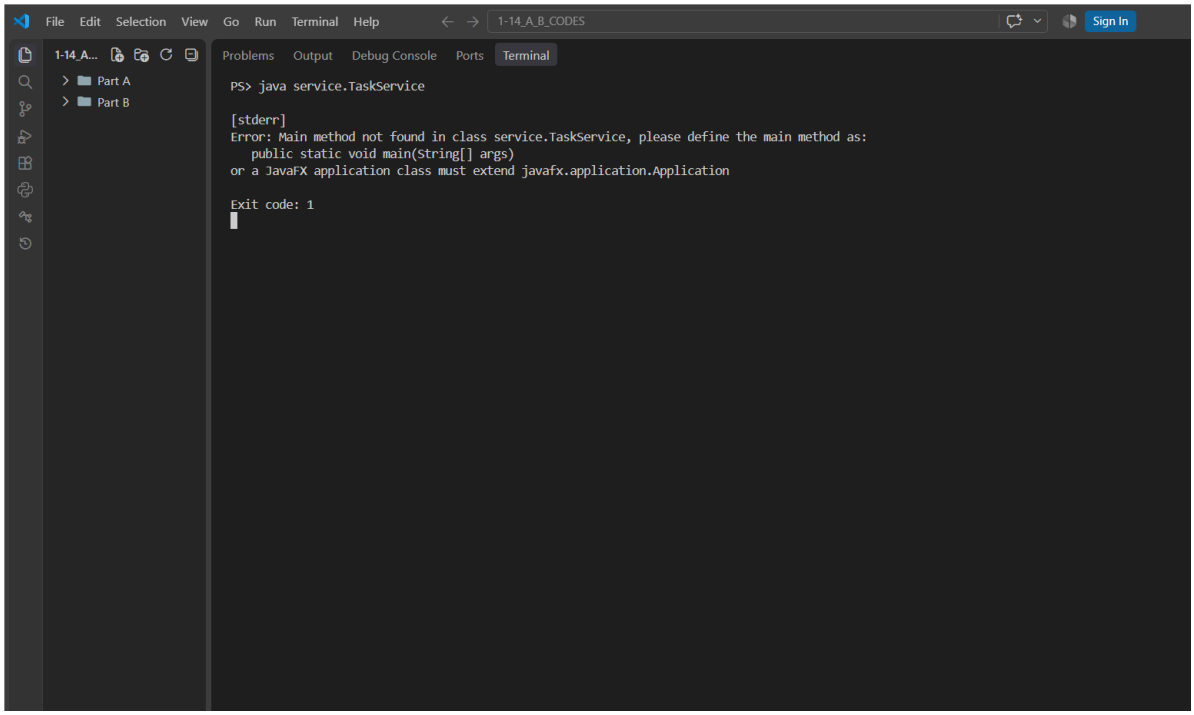
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

The student can assemble a small layered Java application, write basic JUnit tests for expected behaviour and exceptions, and understand how packaging and persistence tests support reliable application delivery.